

A heuristic algorithm based on beam search and iterated local search for the maritime inventory routing problem

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ABSTRACT

The Maritime Inventory Routing Problem (MIRP) is a challenging optimization problem, and there are still no well-established methodologies capable of efficiently solving large instances or their variants due to its high structural complexity. To address this gap, a heuristic framework is proposed for the deterministic, finite-horizon, single-product MIRP. Unlike most existing heuristic approaches, which depend on mathematical programming models and commercial solvers, the presented method integrates Beam Search (BS) and Iterated Local Search (ILS) within an open-source environment. Its contribution lies not in the individual use of BS or ILS, but in their tailored adaptation and combined design, which captures the interdependence between routing and inventory decisions. The methodology introduces a greedy randomized algorithm for evaluating partial solutions during BS, a problem-specific solution representation, and new neighborhoods designed to explore the MIRP solution space efficiently. To the best of our knowledge, this is the first fully self-contained heuristic approach capable of solving all 72 MIRPLib Group 2 instances, a publicly available benchmark dataset, without relying on mathematical programming. It improves the best-known solutions for 18 instances, achieving an average best gap of -1.85% , and delivers competitive performance on the remaining 54, with an average best gap of 2.72% , all with consistent execution times. These results highlight the potential of the proposed heuristic to address large and complex MIRPs and support practical applications in maritime logistics planning.

1. Introduction

In the international shipping industry, operating companies often have to manage product transport within their fleet, respecting inventory levels in ports and vessels in an integrated way. The combination of these elements takes into account the route and scheduling of the ships with inventory management, known in the literature as the *Maritime Inventory Routing Problem* (MIRP) (Christiansen and Fagerholt, 2009). In this challenging combinatorial problem, the routing and scheduling steps define which ports will be visited by each vessel, in which sequence and time. Meanwhile, the inventory management step determines the inventory levels of each port and vessel at each point in time within a given planning horizon (Christiansen and Fagerholt, 2009; da Costa, 2018).

Solving MIRPs is a challenging task due to its alignment with real-world complexities, such as variable demand rates, multi-product

logistics, and the coordination of a central supplier with multiple consumers. Fagerholt et al. (2023) reviewed MIRP research from the past decade and proposed several future directions, including applications in energy transport, such as hydrogen, ammonia, LNG, and liquefied CO₂, considering increasingly aggressive decarbonization goals. They also emphasize a key consideration when addressing this type of problem: the different planning levels, particularly between the tactical and operational levels. Strategic and tactical planning problems typically involve decisions with long-term impacts, while operational planning focuses on planning over shorter time horizons, ranging from a few days or weeks to a few months, and tends to involve much harder constraints. In addition, operational planning problems require tools capable of providing high-quality results in acceptable time frames. As a result, exact methods are generally not favored as they typically deliver optimal solutions only for small-scale instances of practical

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size. Another challenge involving MIRP is the strong interdependence between routing decisions, where factors such as inventory levels and berth limitations trigger chain reactions throughout the supply chain.

Over the past two decades, matheuristics have emerged as a growing field in Operations Research. These are optimization algorithms that combine metaheuristics with mathematical programming techniques (Papageorgiou et al., 2018), often used in the MIRP literature. On the other hand, pure heuristics (i.e. those that do not rely on mathematical optimization procedures) are less commonly applied due to the very constrained nature of the problem, as just building an initial solution can be very challenging (Frisker et al., 2022).

There is no single and well-defined definition of an MIRP, but rather numerous variants tailored to specific industrial applications (Papageorgiou et al., 2014; Andersson et al., 2010). Given this diversity, standardizing the comparison of solution methods and establishing a taxonomy is essential. To address this issue, Papageorgiou et al. (2014) introduced the MIRPLib, a collection of publicly available single-product MIRP benchmark instances. A subset of these instances, called Group 2, serves as the focus of this study. These instances feature a planning horizon of over 120 periods, with no split pickups or deliveries. The problem setting includes a single production port, multiple consumer ports, and various classes of vessels.

In contrast to most existing heuristic approaches for MIRPs, which often rely on mathematical programming models whose efficiency is highly dependent on commercial solvers, in this work we propose an open-source approach that combines Beam Search (BS) (Lowerre, 1976; Ow and Morton, 1988; Reddy, 1977) with Iterated Local Search (ILS) (Lourenço et al., 2003) (BS-ILS) to tackle MIRPLib Group 2 instances. These instances feature a single production port, multiple consumer ports with inventory constraints, and multiple class of vessels, without split pickups or deliveries, creating a highly constrained decision structure between routing and inventory. Evaluating solutions in this context is computationally complex and not straightforward, making the design of effective heuristics particularly challenging. The novelty of our contribution lies not in the isolated use of BS or ILS, but in the way these components are adapted and integrated to address a highly constrained MIRP within reasonable execution times. In addition, our approach addresses the challenge of evaluating partial solutions, proposing an alternative strategy that employs a Greedy Randomized Algorithm (GRA) to evaluate solutions during BS. The article also introduces problem-specific improvements in the local search phase, including a novel neighborhood designed to explore the solution space in ways that are both effective and computationally efficient. Finally, we introduce an innovative solution representation for MIRP, as well as a Julia package designed to support the applicability of the MIRPLib.

To our knowledge, this is the first fully self-contained heuristic method applied and evaluated across all 72 instances of MIRPLib Group 2, offering a practical alternative that achieves high-quality results in competitive execution times. Our goal is to demonstrate the potential of this domain-tailored heuristic framework to solve complex MIRPs in real-world settings.

This paper is organized as follows. Section 2 briefly reviews the literature on deterministic MIRP works and its variants, and presents heuristic applications. Section 3 describes the deterministic MIRP considered in this paper. Section 4 presents the proposed BS-ILS heuristic. Section 5 reports and discusses the results of computational experiments. Finally, Section 6 concludes the work and provides directions for future research.

2. Literature review

2.1. Maritime inventory routing problem

The MIRP refers to the strategic coordination of maritime transportation and inventory management, integrating logistics decisions

with dynamic inventory levels at both origins and destinations. It requires planning vessel routes, determining port call sequences and deciding product quantities for loading and unloading, while ensuring that inventory limits are respected (Sanghikian et al., 2021).

Real-world MIRP applications are often complex, incorporating variations such as supplier/producer or central consumer configurations, inventory constraints at specific ports, variable production/consumption rates, multiple products, and spot charters. To support the development and comparison of solution methods for these variants, standardized instances and modeling frameworks are essential. In this context, Papageorgiou et al. (2014) introduced the MIRPLib to standardize MIRP definitions and support advanced formulations. They proposed a core model for single-product distribution, ensuring inventory constraints at all ports. Building on this foundation, additional MIRP variants were developed, including Group 2 instances — deterministic, long-horizon problems with one loading port and multiple discharge ports. This simplified model assumes that the vessels travel fully loaded to discharge regions and return empty. Detailed descriptions of the original Group 2 model are provided in Section 3.

2.2. Heuristic approaches

Many authors have developed methods to address MIRP, including mathematical models, heuristics, metaheuristics, and matheuristics. Shaabani et al. (2023) and Fagerholt et al. (2023) provided a comprehensive review of the literature on MIRP and the algorithms used. They summarized and classified recent MIRP studies, emphasizing factors such as the size of the problem (e.g., the number of ports, vessels, products, and planning horizon) and the techniques applied to solve them. Out of the 34 studies listed by Shaabani et al. (2023) and the 53 studies listed by Fagerholt et al. (2023), 38 focus on MIRP solutions using heuristic techniques.

Ronen (2002) pioneered this area by introducing a cost-based heuristic. A decade later, Stålhane et al. (2012) proposed a multi-start local search heuristic for the LNG inventory routing problem, using greedy insertion for initialization and refining solutions via neighborhood search, branch-and-bound, or their combination. Uguen et al. (2013) implemented a fix-and-relax time decomposition heuristic for MIRPs with a single product, balancing solution quality with computational efficiency, and Goel et al. (2015) proposed an iterative heuristic for LNG ship scheduling and inventory management.

Metaheuristic approaches have become important for exploring large search spaces. Christiansen et al. (2011) integrated a construction heuristic with a Genetic Algorithm (GA) for multi-product MIRPs. Siswanto et al. (2011a) proposed a multi-heuristic methodology that achieved optimality for smaller instances. Expanding further, Siswanto et al. (2011b) developed a multi-heuristic strategy for a ship inventory routing and scheduling problem with non-dedicated compartments. More recently, Siswanto et al. (2019) proposed a GA based on multi-heuristics for an MIRP with multiple time windows. Sheikhtajian et al. (2020) studied uncertain LNG marine IRPs and proposed a meta-heuristic that combines heuristics, GA, and fuzzy modeling to improve sensitivity analysis.

Hybrid matheuristics have also been explored. Chandra et al. (2013) combined Ant Colony Optimization with linear programming for MIRPs with optional cargoes, while Song and Furman (2013) applied a large neighborhood search (LNS) heuristic to iteratively solve subproblems. De et al. (2017) applied Particle Swarm Optimization (PSO) to sustainable MIRPs with time windows, while Zhang et al. (2018) used a Lagrangian heuristic to manage delivery time uncertainty. Al-Haidous et al. (2022) investigated a short-term LNG scheduling problem, analyzing transportation expenses and emissions using PSO.

Matheuristics, which integrate mathematical programming and heuristics, are a powerful tool for solving MIRPs. Agra et al. (2014) introduced hybrid heuristics for short-sea inventory routing, combining

rolling-horizon heuristics (RHH) with local branching and feasibility pumps. Shao et al. (2015) proposed a two-stage heuristic for LNG-IRPs, integrating a RHH with MIP-based neighborhood refinement methods. Hemmati et al. (2015, 2016) developed hybrid matheuristics for cargo and inventory routing in tramp shipping. Meanwhile, Mutlu et al. (2016) focused on the LNG Annual Delivery Program (LNG-ADP), which is a schedule of LNG deliveries for a given year, introducing a matheuristic with a unique vessel selection methodology.

Agra et al. (2016) developed a MIP-based heuristic with local search techniques, while Agra et al. (2017) proposed a matheuristic to improve solutions and ensure feasibility when the mathematical model fails. Agra et al. (2018) and Diz et al. (2019) refined MIRP decomposition algorithms using an Iterated Local Search (ILS) heuristic and robust optimization. Sanghikian et al. (2021) designed a variable neighborhood search (VNS) for a multi-product MIRP, integrating LP solvers to optimize continuous variables for routing solutions.

Recent research has focused on hybrid methods, parallel strategies, and uncertainty modeling. Dong et al. (2018) developed a RHH re-optimization framework for combined maritime fleet deployment and inventory management with port-visit flexibility in roll-on/roll-off shipping. Stanzani et al. (2018) proposed a multi-start heuristic for the simultaneous pickup and delivery ship routing, incorporating two enhancement methods and a local search for further solution improvement. Furthermore, Msakni and Haouari (2018) applied an optimization-based VNS for short-term LNG delivery planning, using a greedy heuristic for initialization and iteratively solving reduced instances with a commercial solver. For LNG-ADPs problems, Li and Schütz (2020) proposed an RHH, while Li et al. (2023) incorporated transshipment and intermediate storage, using various RHH configurations.

In MIRPLib-specific studies, Papageorgiou et al. (2018) introduced several heuristics, such as RHH, K -opt heuristics, local branching, solution polishing and hybrid approaches, that combine metaheuristics with mathematical programming to solve MIRPLib's Group 2 instances. Their study resulted in new best-known solutions for 26 out of 70 instances that had not yet been proven optimal, and they achieved new best-known lower bounds for 56 instances. Munguía et al. (2019) presented the Parallel Alternating Criteria Search (PACS), an LNS-based heuristic that yielded strong results for Groups 1 and 2.

Friske and Buriol (2020) implemented a multi-start heuristic with LNS for Group 1 instances, while Friske et al. (2022) employed a relax-and-fix algorithm followed by fix-and-optimize to refine MIRPLib's Group 1 solutions. Wang et al. (2023) introduced a bilevel parallel memetic algorithm for a maritime location inventory routing problem, tested in Group 1 instances, achieving near-best solutions with favorable deviations and faster computation times. Table 1 summarizes the methods described in this section.

In contrast to the reviewed studies, which often rely on mathematical models dependent on commercial solvers, this study employs a heuristic approach that does not require third-party solvers, distinguishing itself through accessibility and applicability to real-world scenarios. Additionally, a very limited number of works address the full set of MIRPLib Group 2 instances. A summary of how the proposed approach compares with previous work is provided in Table 2.

3. Problem description

The formulation presented in this section corresponds to the MIP model for Group 2 instances available in MIRPLib (Papageorgiou, 2013). The Group 2 instances describe long-horizon deterministic single product maritime inventory routing problems. In the following description, the terms of the sets “port” (J) and “region” (R) are used as synonyms, since it is assumed that there is one port per region.

In a defined time horizon T , a heterogeneous fleet of ships, classified into different vessel classes VC , each with distinct capacities Q^{vc} , is assigned to service the production ports J^P and the consumption ports

J^C . This operation must be done while ensuring that the minimum and maximum inventory limits in port $j \in J$ ($S_{j,t}^{min}$ and $S_{j,t}^{max}$, respectively) are respected. Each port $j \in J$ is characterized by an initial inventory $s_{j,0}$ and a specific production or consumption rate $d_{j,t}$ in period $t \in T$, along with an associated indicator parameter Δ_j , which takes the value 1 for production ports and -1 for consumption ports. Berth limit (B_j) constraints must also be respected, restricting the number of vessels that can attempt to load or discharge in a port $j \in J$ in a given period of time. In this MIRP, a vessel must travel at full capacity from a loading port to a discharging port, and empty from a discharging port to a loading port.

The formulation follows an arc-flow mixed-integer linear programming model, where a set of nodes $\mathcal{N}$ represents the port-time pairs. A set of arcs A^{vc} associated with each vessel class $vc \in VC$ defines the possible routes between nodes. Furthermore, a set of nodes $\mathcal{N}_{s,t}$ is defined, including both the source n_s and the sink n_t nodes. The set of inter-regional arcs $A^{vc,ir}$ represents the arcs between the production and consumption regions. For each vessel class $vc \in VC$ and for each node $n \in \mathcal{N}_{s,t}$, the incoming and outgoing arcs are denoted by RS_n^{vc} (for reverse star) and FS_n^{vc} (for forward star), respectively, while the set of forward star inter-regional arcs is denoted by $(FS_n^{vc,ir})$.

All routing costs are incorporated into C_a^{vc} (cost for vessel of vessel class $vc \in VC$ to traverse the arc $a \in A^{vc}$). Arc costs may vary depending on the type of arc (source, sink, waiting, inter-port). The distance δ_{j_1,j_2} between ports j_1 and j_2 , the cost per kilometer per vessel C_{km}^{vc} according to its vessel class $vc \in VC$ and the fees π_j of each port $j \in J$ are considered. Some discounts are also included, a reward for finishing the route before the time horizon ends ρ and a discount γ^{vc} for vessels of a vessel class $vc \in VC$ traveling empty from a consumption port to a production port. Inventory costs $P_{j,t}$ are included as penalties, reflecting production and consumption in each port $j \in J$ to comply with inventory constraints in each period $t \in T$.

Integer decision variables x_a^{vc} represent the number of vessels in a given vessel class $vc \in VC$ that traverse a specific arc $a \in A^{vc}$. The continuous decision variables $\alpha_{j,t}$, representing a spot charter, represent the amount to be produced or consumed in port $j \in J$ during period $t \in T$, and $s_{j,t}$ the quantity of inventory units in port $j \in J$ at the end of period $t \in T$.

All indices, sets, parameters, and decision variables are detailed in Appendix A. The MIRP model (Papageorgiou, 2013) is described as follows:

$$\begin{aligned} \max z = & \sum_{vc \in VC} \sum_{a \in A^{vc}} -C_a^{vc} x_a^{vc} && \text{(routing cost)} \\ & + \sum_{j \in J} \sum_{t \in T} -P_{j,t} \alpha_{j,t} && \text{(inventory penalties cost)} \end{aligned} \quad (1)$$

subject to

$$\begin{aligned} & \sum_{a \in FS_n^{vc}} x_a^{vc} - \sum_{a \in RS_n^{vc}} x_a^{vc} \\ & = \begin{cases} 1 & \text{if } n = n_s, \\ -1 & \text{if } n = n_t, \\ 0 & \text{if } n \in \mathcal{N} \setminus \{n_s, n_t\} \end{cases} \quad n \in \mathcal{N}_{s,t}, vc \in VC \end{aligned} \quad (2)$$

$$\begin{aligned} s_{j,t} &= s_{j,t-1} \\ & + \Delta_j \left(d_{j,t} - \sum_{vc \in VC} \sum_{a \in FS_n^{vc,ir}} Q^{vc} x_a^{vc} - \alpha_{j,t} \right) \quad n = (j, t) \in \mathcal{N} \end{aligned} \quad (3)$$

$$S_{j,t}^{min} \leq s_{j,t} \leq S_{j,t}^{max} \quad n = (j, t) \in \mathcal{N} \quad (4)$$

$$\sum_{vc \in VC} \sum_{a \in FS_n^{vc,ir}} x_a^{vc} \leq B_j \quad n = (j, t) \in \mathcal{N} \quad (5)$$

$$\alpha_{j,t} \geq 0 \quad n = (j, t) \in \mathcal{N} \quad (6)$$

$$x_a^{vc} \in \{0, 1\} \quad vc \in VC, a \in A^{vc,ir} \quad (7)$$

Table 1

Overview of MIRP heuristics methods.

Reference	Application	Method
Ronen (2002)	Refined multi-product	Cost-based
Christiansen et al. (2011)	Cement multi-product	Genetic Algorithm
Siswanto et al. (2011a,b)	Refined multi-product	Genetic Algorithm
Stålthane et al. (2012)	LNG	Multi-start local search
Uggen et al. (2013)	LNG	Fix-and-relax time decomposition
Chandra et al. (2013)	Automobiles multi-product	Ant Colony Optimization
Song and Furman (2013)	Bulk single-Product	Large neighborhood search
Agra et al. (2014)	Refined multi-products	Rolling-horizon, local branching, and feasibility pumps
Goel et al. (2015)	LNG	Iterative heuristic search algorithm
Shao et al. (2015)	LNG	Rolling-horizon with MIP-based neighborhood refinement methods
Agra et al. (2016, 2017, 2018)	General and Fish Feed single-product	MIP-based heuristic employing local search techniques
Hemmati et al. (2015, 2016)	General single and multi-product	Adaptive large neighborhood search and mathematical programming
Mutlu et al. (2016)	LNG	Matheuristic with a unique vessel selection methodology
De et al. (2017)	Containers single-product	Particle Swarm Optimization
Dong et al. (2018)	General single-product	Rolling-horizon re-optimization framework
Msakni and Haouari (2018)	LNG	Variable neighborhood search
Papageorgiou et al. (2018)	MIRPLib Group 1 and 2	Rolling horizon heuristics, K-opt heuristics, local branching, solution polishing, and hybrid approaches, integrated with a matheuristic approach
Stanzani et al. (2018)	Crude-oil	Multi-start heuristic
Zhang et al. (2018)	LNG	Lagrangian heuristic
Diz et al. (2019)	Crude-oil	Combined relax-and-fix and fix-and-optimize
Munguía et al. (2019)	MIRPLib Group 1 and 2	Parallel Alternating Criteria Search, a heuristic based on large neighborhood search
Friske and Buriol (2020)	MIRPLib Group 1	Multi-start heuristic with large neighborhood search
Li and Schütz (2020)	LNG	Rolling-horizon heuristic
Sanghikian et al. (2021)	Refined multi-product	Variable neighborhood search integrating LP solvers
Sheikhtajian et al. (2020)	LNG	Matheuristic combining heuristics, Genetic Algorithms, and fuzzy modeling
Al-Haidous et al. (2022)	LNG	Particle Swarm Optimization
Friske et al. (2022)	MIRPLib Group 1	Relax-and-fix algorithm followed by fix-and-optimize
Li et al. (2023)	LNG	Rolling-horizon heuristic
Wang et al. (2023)	MIRPLib Group 1	Bi-level parallel memetic algorithm

Table 2

Works that specifically addressed MIRPLib Group 2 instances.

Author	Solver-based	Key features
Papageorgiou et al. (2018)	Yes	Metaheuristics with mathematical programming (RHH, K-opt heuristics, local branching)
Munguía et al. (2019)	Yes	PACS, an LNS-based heuristic
This paper	No	BS-ILS

$$x_a^{vc} \in \mathbb{Z}_+ \quad vc \in VC, a \in A^{vc} \setminus A^{vc,ir}. \quad (8)$$

In this model, the objective function (1) minimizes the sum of transportation costs, and penalties for lost production and stockout. Constraints (2) ensure the flow balance of vessels within each vessel class. The minimum and maximum inventory limits at ports must be respected, while also ensuring the continuity of inventory flow, as defined by Constraints (3) and (4), respectively. Constraints (5) represent the berth limit constraints. Constraints (6) ensure that the spot charter variable is always non-negative. Constraints (7) ensure that at most one vessel per vessel class travels along a particular interregional arc. Finally, Constraints (8) define the variables' domains for the remaining arcs.

Solving MIRP using exact mathematical models is often impractical due to their high computational cost. As the problem size increases, particularly in real-world instances, solution times become unmanageable, limiting the applicability of exact approaches. Consequently, exact methods are limited to small or simplified cases, highlighting the need for scalable alternatives like heuristics or metaheuristics that can provide high-quality solutions within a reasonable execution time.

4. Proposed heuristic

The developed heuristic is a modified version of Beam Search (Lowerre, 1976; Ow and Morton, 1988; Reddy, 1977), which serves as a constructive algorithm, complemented by local search procedures that improve all the generated solutions. Once these are concluded, an Iterated Local Search (Lourenço et al., 2003) is applied to possibly refine the incumbent solution.

A high-level overview of the algorithm is presented in Fig. 1. The process begins with a modified version of the BS heuristic which explores a limited set of the most promising solution paths at each step. As further detailed in Section 4.3, instead of expanding all possible successors, as in a traditional breadth-first search, it maintains only a fixed number of the best candidates, reducing computational effort while directing the search towards high-quality solutions, shown as triangles. These solutions undergo a local search (LS) improvement based on a Randomized Variable Neighborhood Descent (RVND) procedure (Subramanian et al., 2010), indicated as circles. Finally, the incumbent solution is possibly further refined by the ILS approach, yielding the final solution, presented in diamond form.

4.1. Solution representation

The solutions are represented by a single vector that stores port-vessel information, defined as a *call*, and the order of the calls in the vector corresponds to the chronological order of the ships' visits to these ports. For a call to occur, the preceding ones must already have taken place, with respect to the port and the vessel. Thus, a ship requiring to service a call in a port must wait its turn in the list to perform the operation. The sequence is chronological, but only when it involves the same ports or vessels across the calls. For example,

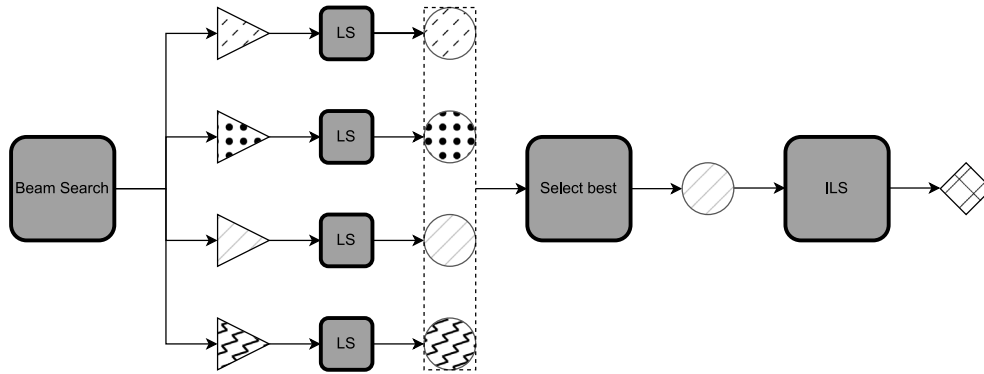


Fig. 1. Schematic representation of the proposed algorithm integrating BS with LS and ILS components. The BS provides an initial set of solutions, which are refined by the LS procedure. The best solution is selected and then the ILS iteratively perturbs and improves the solution to escape local optima and enhance solution quality.

if the first port to be serviced is port P1 by vessel V3, followed by port P1 again, but this time by vessel V1, vessel V3 must end its operations at port P1 before vessel V1 can begin (considering that there is only one berth available at port P1). On the other hand, if the next call is unrelated, it may occur prior to the last one. To ensure feasibility, vessels must always travel between a production port and a consumption port, in either direction. This rule prevents vessels from traveling between ports of the same type, ensuring operational consistency.

This solution representation allows a given solution to have multiple representations, leading to redundancy. Two solutions can correspond to the same actual solution, as the calls involve distinct ships servicing different ports. For example, a solution may evaluate (P1,V1) followed by (P3,V2), while another solution evaluates (P3,V2) first and then (P1,V1). Despite the difference in visit order, both solutions correspond to the same operation in real life. This illustrates that permutations may not affect the final outcome and can be considered equivalent in the problem context.

Note that it is not straightforward to check which solutions are equivalent. A specific approach has been created to address this issue with some auxiliary structures as follows. For each element in the vector, four auxiliary pointers are maintained, each storing the position of the last occurrence of the vessel, the last occurrence of the port, the next occurrence of the vessel, and the next occurrence of the port. This information is preprocessed and used during local search procedures to restrict their moves to non-redundant operations, avoiding unnecessary computation. In addition, the time for each vessel and port is also preprocessed to speed up the solution evaluation.

4.2. Solution evaluation

With the aim of evaluating a solution, each call keeps key information such as the most recent service time at the respective port, the last service time of the vessel, inventory levels, the time of the next violation and the accumulated solution costs (including both routing and inventory components), in addition to the previously described pointers. Depending on the type of change introduced, whether by local search, ILS perturbation, or the greedy randomized algorithm itself, any change in routing leads to the evaluation, responsible for recalculating the relevant solution information.

Upon evaluation of a call, the arrival and operation times of the vessel in the port are calculated considering berth constraints and avoiding potential infeasibility. If the operation occurs within acceptable limits and no constraint is violated, then no penalty is incurred, and inventory

levels at the ports are updated accordingly. However, if a violation occurs prior to the scheduled operation, penalty costs are applied, and inventory levels are corrected to the admissible limit (upper or lower), depending on the port type (production or consumption).

Subsequently, routing costs are calculated and the planning horizon is checked. Calls exceeding the horizon are discarded, and the final inventory violations and total routing costs are recalculated. At the end of each evaluation, all relevant information is updated to ensure consistency, allowing for the next evaluation in upcoming movements.

4.3. Beam search algorithm

The Harpy Speech Recognition System (Lowerre, 1976) was the first to implement what later became known as Beam Search (Ow and Morton, 1988), introducing a heuristic-based approach to efficiently explore and prune the search space in speech recognition tasks. Although the procedure was originally referred to as the Locus Model of Search, the term Beam Search was already used by Reddy (1977). BS is a method for exploring decision trees, especially when dealing with a large solution space. It involves methodically expanding a limited set of solutions simultaneously, aiming to increase the chance of finding a high-quality solution with limited searching effort.

The BS algorithm, as described in Morais et al. (2024), is derived from the concept of breadth-first search. It explores a graph by expanding the most promising nodes while maintaining a fixed number of nodes at each level, as shown in Fig. 2. The approach is structured into six stages: Initialization, Node Expansion, Successor Evaluation, Beam Selection, Iteration, and Termination and Output generation.

During the Initialization stage, the algorithm establishes the starting parameters, including the initial node, which will always be an empty list, alongside the beam width. In the Node Expansion phase, successors are formed for each active node. Each successor is a partial solution. For a given node and its corresponding solution, all valid calls that can be appended to the solution are considered. Then, for every call, a new node is created that incorporates all previous calls together with the new one. Consequently, at the i th level of the generated tree, every solution consists of i calls. In the Successor Evaluation phase, a Greedy Randomized Algorithm (GRA) is employed to assess the quality of each successor. Next, the Beam Selection phase involves selecting the most promising successors based on the heuristic evaluations while adhering to the beam width limit. The process continues with the Iteration stage, repeating until no further successors can be generated from the final level within the planning horizon. In the Termination and Output generation phase, the best solutions generated in all steps are selected.

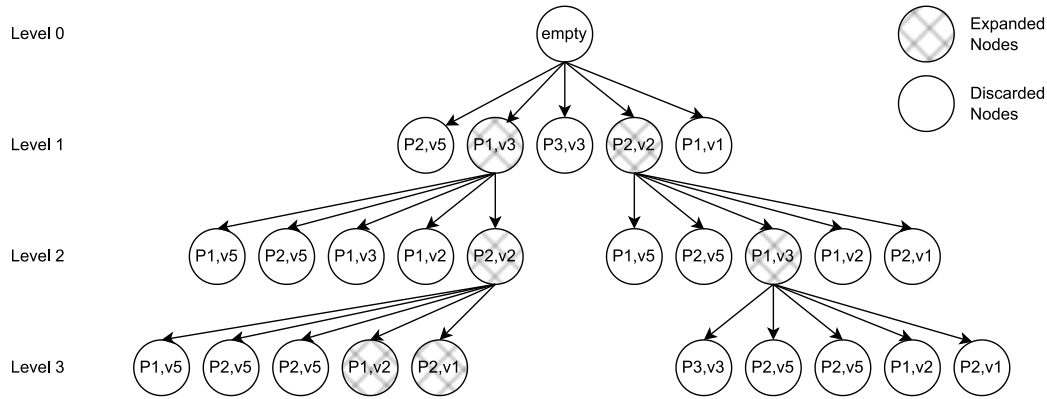


Fig. 2. Schematic representation of the first three levels of Beam Search Algorithm with $w = 2$ and $N = 2$. Each marked node indicates the successor from the specific i th level. The successor is selected by the GRA. The algorithm continues until no further successors can be generated from the final level within the planning horizon.

In the present study, the BS heuristic is configured to accept three different parameters as input, which have an impact on different phases of the BS procedure. The parameters are the number of nodes allowed at a given level (N), the number of greedy solutions for each node (q) and the maximum number of children per node (w).

In the context of the proposed BS algorithm, an increase in the number of evaluated nodes, driven by a deeper or wider search tree, directly impacts execution time. Consequently, the initial parameter to consider is N , which represents the number of nodes allowed within a specified level. The challenge lies in selecting a value that ensures an adequate array of potential initial solutions while keeping the tree's size within reasonable limits, which would otherwise result in higher execution time. This effect becomes even more critical as the problem instance increases in complexity, particularly with the addition of more ports, vessels, or a longer planning horizon. Although a more extensive exploration of the solution space may be acceptable for small instances, it can quickly become computationally prohibitive for larger ones. Therefore, careful calibration of the tree structure is essential to ensure a proper balance between solution quality and computational efficiency.

The Successor Evaluation phase follows the Node Expansion phase. Comparing partial solutions in the MIRP is a high challenge. Therefore, during the evaluation of successors, different criteria were tested. Comparison of incomplete solutions can lead to imprecise and unsatisfactory evaluations. As a result, a GRA is applied to evaluate the quality of each generated successor. Thus, the second parameter, denoted as q , specifies the number of greedy solutions generated during the process. Specifically, the greedy heuristic is executed once following a deterministic approach, and subsequently, it is repeated $q - 1$ times incorporating randomness into its execution to introduce variability in the solution space. To select the best successors, the median of the associated q values is used as a representative measure. The third parameter, denoted as w , represents the upper limit on the number of children that can be generated by a single node. This parameter plays an important role in maintaining the diversity and heterogeneity among the solutions.

One problem already mentioned in Section 4.1 is the fact that several identical solutions are represented in different ways. A simple procedure was devised with the aim of reducing the occurrence of solutions that share the same cost, thereby improving the efficiency of BS, as it is undesirable to occupy unnecessary space in the BS level. This procedure functions by keeping a list in which the keys correspond to the scores of potential solutions. Referencing this list, it is possible to

identify whether a particular score has been assigned to any solution. In such cases, these solutions are excluded from further consideration. An unlikely issue may arise with the presence of two distinct solutions that share the same cost, due to the intricate nature of the problem. Nevertheless, even if this occurs, the benefits continue to outweigh the disadvantages.

4.4. The greedy randomized algorithm

This work innovatively addresses the challenge of evaluating partial solutions by proposing an alternative strategy to assess them during the BS. To effectively compare partial solutions, i.e., the successors produced during a BS execution, a greedy heuristic is used to construct a complete solution starting from a given partial solution, as presented in Algorithm 1. The heuristic builds a feasible solution using a greedy method, making each decision by selecting the most promising immediate choice according to a defined criterion. The criterion used involves choosing the next port at risk of violating inventory levels and identifying which vessel can arrive there first. Therefore, the path is designed to prioritize, in the subsequent call, the inclusion of the port at risk of inventory violation and the vessel capable of reaching this port in the earliest possible time.

Algorithm 1: Greedy Algorithm — Deterministic version

```

1  $s_{partial} \leftarrow BS(instance, time\_horizon, N, w)$ 
2 while  $t \leq time\_horizon$  do
3    $next\_port \leftarrow next\_port\ to\ violate$ 
4    $next\_vessel \leftarrow vessel\ arrives\ first$ 
5    $s_{partial} \leftarrow s_{partial} \cup (next\_port, next\_vessel)$ 
6 end
7  $S \leftarrow s_{partial}$ 
8 return  $S$ 

```

To enhance flexibility and solution exploration, a nondeterministic version of the greedy algorithm is implemented, introducing randomness at two stages of the process, the port selection and the vessel selection. This randomness mechanism allows the algorithm to explore a wider range of possible paths, helping to diversify the solutions generated by the constructive method and potentially increasing the chances of finding better solutions compared to the deterministic version. Instead of choosing the port that first violates its capacity constraint or the ship that first serves the selected port, ports with earlier violation

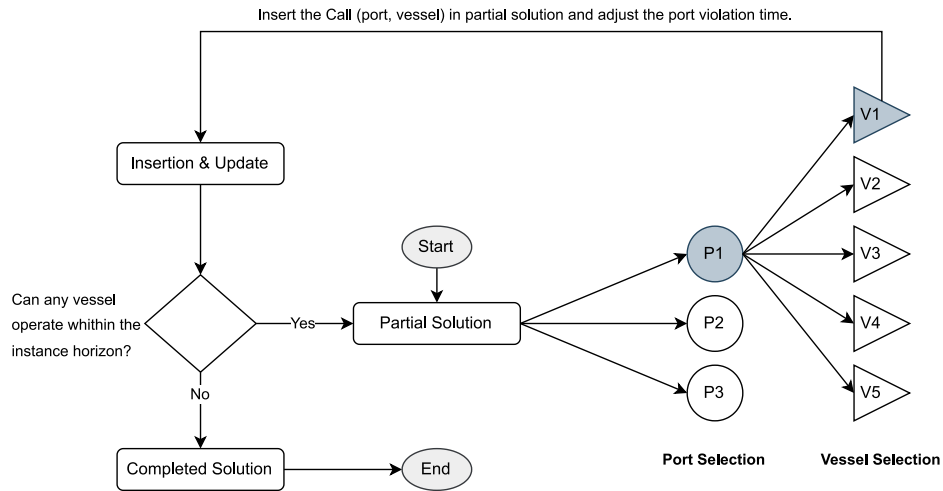


Fig. 3. Schematic representation of the GRA. Each partial solution is evaluated using a completion procedure. At each step, a port is selected, and then the vessel with the earliest arrival time is chosen. The corresponding call is added to the solution, and this process continues until the planning horizon is reached. The resulting complete solution is then used to assess which node is the most promising.

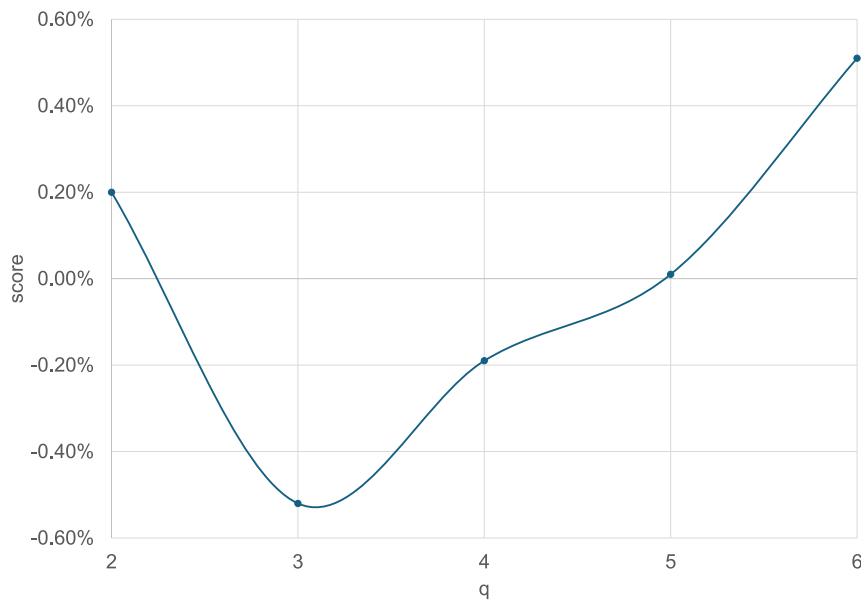


Fig. 4. Calibration of greedy solutions per node (q) under fixed execution time by inversely adjusting nodes per level (N), showing average objective gaps on easy instances ($N = 100$, 10 seeds).

times and vessels with earlier arrival times have a higher probability of being selected.

By incorporating a probability factor within a normal distribution and considering the standard deviation, it is possible that the next port on the route might not be the first to face inventory violation, and the selected vessel may not be the one that reaches there first. The normal distribution was used to sample values around the average, introducing variation but ensuring that the majority of samples close to that remained near the central value.

Notably, throughout the BS procedure, multiple complete solutions are built using this GRA. To ensure that valuable solutions are kept, we keep the best N solutions identified by the greedy heuristic, in case

they outperform the final solution of the BS. Fig. 3 presents a schematic representation of the GRA.

4.5. Local search

The local search procedure is based on a Randomized Variable Neighborhood Descent (RVND) scheme, where each neighborhood is explored in sequence until an improvement is found. If no improvement is found, the procedure moves to the next neighborhood, following a random order. If any neighborhood results in a better solution, the process is restarted. Furthermore, our implementation uses the first improvement strategy.

An important consideration when considering these neighborhoods is that in some cases certain calls may be displaced outside the planning

horizon. When this issue occurs, these calls are ignored during the solution evaluation, as they do not contribute to the cost or feasibility within the planned period.

Due to the challenges of applying neighborhoods in the MIRP, several options were tested, including a novel neighborhood designed to explore the solution space in a manner that is both effective and computationally efficient. In the end, six neighborhood structures proved to be the most effective and are implemented as described below.

- **Swap:** swap two *calls* in the solution.
- **Relocate:** a *call* is relocated to a new position in the solution.
- **Replace:** the port of a *call* is replaced by a port of the same type.
- **Insert:** a pair of *calls* is inserted at the end of the solution.
- **Remove:** removes a *call* along with the next *call* of the same vessel. This procedure is carried out to preserve vessel parity. For example, if the *call* (P3, V1) is removed, it becomes necessary to remove the next call of V1 to maintain the producer–consumer balance.
- **Swap Port:** the destinations of two vessels are swapped, while restricting the swap to ports of the same type. Its strategy is specific to the MIRP and directly modifies ships' routes, with no risk of breaking parity and no redundancy, as every route modification produces a different solution.

4.6. Iterated local search

ILS (Lourenço et al., 2003) applies an iterative process that alternates between perturbation and local search procedures. In each iteration, a perturbation is randomly applied to the current solution, which involves making a controlled change in its structure, by altering the routing structure of the current solution, triggering a change in the cost value of the solution. The perturbation mechanism selects one of the six neighborhoods defined in Section 4.5 and applies a random move. In our implementation, such a mechanism is called twice in a row. This perturbation generates a new candidate solution which is then refined using a local search to improve its quality within the neighborhood of the current solution. The algorithm then checks if the new solution will be accepted by a pre-established Simulated Annealing criterion (Kirkpatrick et al., 1983). Once a solution is accepted, the algorithm updates the current solution and checks for improvement. If no improvement occurs, a counter increments. When this counter exceeds the allowed number of non-improving iterations, the solution is restored, and the counter resets. This process repeats until the maximum number of iterations is reached, at which point the final solution is returned.

5. Computational experiments

The proposed algorithm was coded in Julia 1.10.5 and executed on an AMD EPYC 9654 (Zen 4) @ 2.40 GHz running Gentoo Linux 2.14 limited to a single thread and 16 GB of RAM.

5.1. Instance details

Benchmark instances and their best-known solutions are maintained on the MIRPLib website.¹ The metadata includes high-level information for each instance. Most of the parameters are self-explanatory and were previously presented in Section 3. The reader is referred to Papageorgiou et al. (2014) for further explanation. A software package has been created to read and use these public instances.²

This paper addresses Group 2 instances, representing a single-product MIRP, considering only one port per region and never involving

Table 3

Average and median objective gaps (%) for maximum children per node parameter (w) on easy instances ($N = 100$, 10 seeds).

w	2	3	4	5	6	7
Average	−0.70%	−0.44%	1.01%	0.60%	−0.21%	−0.26%
Median	−0.13%	−0.01%	0.05%	0.00%	0.00%	0.00%

split pickups or deliveries. All instances have one loading region (LR) and different numbers of discharging regions (DR). For example, instance LR1DR02VC01V6a has one loading region, two discharging regions, six vessels of one class, while instance LR1DR04VC03V15b has one loading region, four discharging regions, 15 vessels of three classes. The final letter “a” or “b” refers to the production/demand rates in the ports. If the letter is “a” the rates are constant, and if it is “b” the rates vary during the time horizon.

According to the classification presented by Papageorgiou et al. (2018), instances are grouped into three distinct categories based on their difficulty: Easy (E), Medium (M), and Hard (H). An instance is categorized as hard if, within a time limit of 1800 s, no commercial MILP solver operating with default parameter settings obtains a reduction greater than 10% of the gap. In contrast, an instance is considered easy if at least one commercially available MILP solver can reduce the gap by more than 90% within the same time limit using the default solver settings. Instances that do not meet the criteria for the easy or hard categories are classified as medium.

5.2. Parameter tuning

5.2.1. BS parameters

To establish a promising value for the maximum number of children per node w , a set of experiments was carried out. The tests were carried out on easy instances, with $N = 100$ and 10 seeds per instance. Table 3 reports the average and median objective value gaps obtained from tests in which the parameter w was changed from 2 to 7. The percentages represent the aggregated gap in the objective values across all runs. The best performance was achieved with $w = 2$, which resulted in the lowest average and median gaps.

5.2.2. GRA parameters

Further tests were carried out to obtain the number of greedy solutions q for each node, with the aim of maximizing performance relative to execution time. To this end, the number of nodes allowed per level in the BS N was set to be inversely proportional to q , ensuring that the execution time remained approximately constant. Fig. 4 illustrates the results, showing the calibration of q for easy instances. The score indicates the average gap obtained across 10 different seeds. The best results were achieved for $q = 3$, as it yielded the smallest average gap.

5.2.3. ILS parameters

To tune the ILS parameters, the Julia HyperTuning.jl package (de Dios et al., 2021) was used for automated hyperparameter optimization. We selected 15 instances and five seeds, conducting a total of 120 trials that lasted 72 h. The parameter range, step, and results are detailed in Table 4. The results indicated the final set of parameter values that corresponds to the configuration that achieved the best overall performance.

5.3. Results analysis

5.3.1. Proposed heuristic results

Two studies (Papageorgiou, 2013; Papageorgiou et al., 2018) used MIRPLib Group 2 instances. In the first study, the authors first performed a warm-up phase to obtain an initial good solution, which was then used to warm-start the solver. Subsequently, the main experiments were carried out for 24 h. In the following research (Papageorgiou

¹ <https://mirplib.scl.gatech.edu>.

² <https://github.com/rafaelmartini/MIRPLib.jl>.

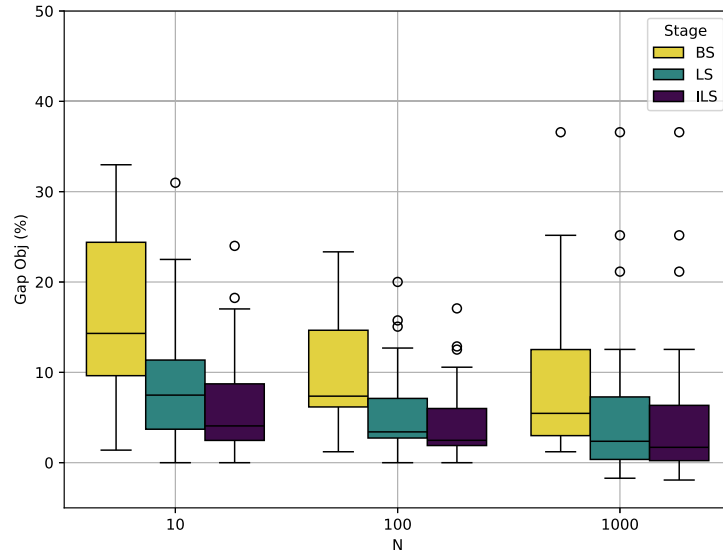


Fig. 5. Boxplot comparing heuristic stages (BS, LS, and ILS) across different N values (10, 100, 1000) on instances with horizon $H = 120$. The y-axis represents objective gap percentages, while the x-axis corresponds to N .

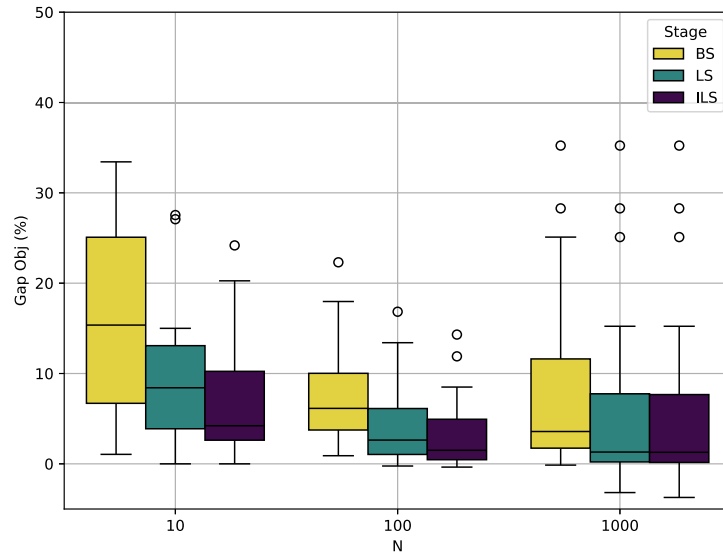


Fig. 6. Boxplot comparing heuristic stages (BS, LS, and ILS) across different N values (10, 100, 1000) on instances with horizon $H = 180$. The y-axis represents objective gap percentages, while the x-axis corresponds to N .

Table 4

ILS hyperparameter optimization. Parameter ranges and steps are detailed, with the final configuration (Result) achieving the best overall performance.

Parameter	Range	Step	Result
SA initial probability	0.75–0.99	0.01	0.79
SA final probability	0.01–0.10	0.01	0.01
ILS number of iterations	10–1000	10	640
ILS number of non-improving iterations before solution restore	1–20	1	4
ILS number of perturbations	1–5	1	2

et al., 2018), the overall execution time per scenario remains unclear, due to the combination of multiple techniques with different times. Moreover, Papageorgiou (2013) emphasizes that it is difficult to find good solutions for Group 2 in a few minutes, establishing a balance between the quality of the results and the time required to obtain them. Therefore, this work aims to evaluate how the heuristic works under different execution times. To accomplish this, the parameter N for the BS is evaluated using three different values ($N \in \{10, 100, 1000\}$). This approach is intended to analyze how results improve with increased execution time using the work by Papageorgiou et al. (2018) as a baseline reference. Another important point is that the gap reported in our results is the best gap obtained from the executed runs in 90,000 s. If any stage of the main algorithm was not completed within this time limit, the reported result corresponds to the last completed stage. For

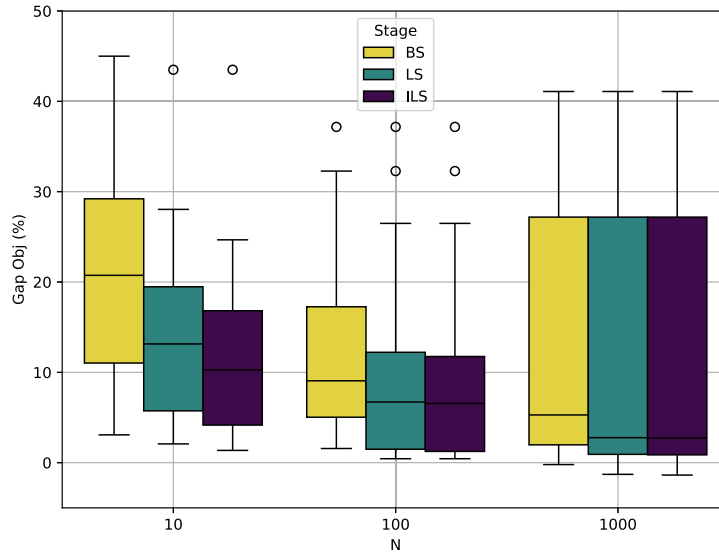


Fig. 7. Boxplot comparing heuristic stages (BS, LS, and ILS) across different N values (10, 100, 1000) on instances with horizon $H = 360$. The y-axis represents objective gap percentages, while the x-axis corresponds to N .

example, if the algorithm does not finish the ILS phase, the reported value reflects the best result achieved up to the LS, along with its respective time. Similarly, if the algorithm does not complete the LS, the reported result comes from the BS, including both the objective value and the time. In instances where the algorithm fails to finish the BS stage, the best result achieved along with the corresponding time are reported.

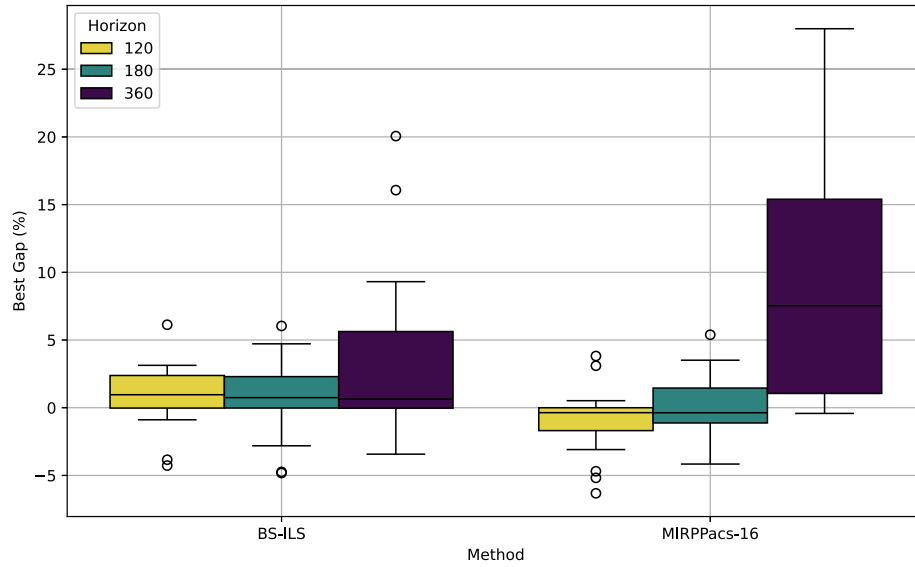
The box plots presented in Figs. 5, 6, and 7 illustrate the comparison between different stages (BS, LS, and ILS) and the variable N over three distinct values (10, 100, and 1000), providing relevant information on the influence of the parameter N on the results. The y-axis represents the percentage of the objective gap (Gap Obj), while the x-axis denotes the different values of N .

A general observation indicates that as the value of N increases from 10 to 1000, there is a visible decrease in the median Gap Obj across all stages, particularly in the BS stage, which is directly affected by this parameter. This suggests that larger values of N are associated with lower percentages of gaps. Furthermore, it can be concluded that while LS has a considerable influence in all the cases analyzed, the effect of ILS tends to decrease as the value of N increases. This observation remains consistent across various instance sizes, with the only notable variation being the increase in Gap Obj for larger instances. Furthermore, the interquartile ranges for each stage illustrate that there is a significant reduction in variability as N increases from $N = 10$ to $N = 100$. At $N = 10$, all stages exhibit a wide spread in values, with ILS showing the smallest variability. At $N = 100$, it can be observed a more consistent performance between stages. However, for $N = 1000$, as the complexity of the instances increases, particularly for those with a horizon of 360, the variability becomes the highest among the cases. This occurs because, with $N = 1000$, the method is unable to complete the heuristics stages within the 90,000 s limit. Therefore, it is preferable to work with a smaller N and complete the entire algorithm rather than increasing N , given that the maximum execution time is constrained.

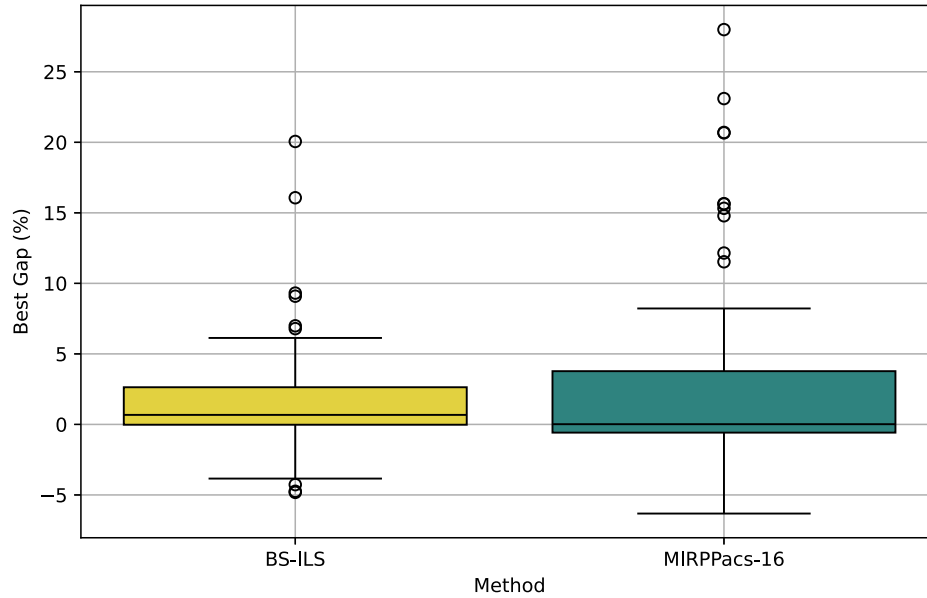
To evaluate the effectiveness of the proposed framework and its heuristic strategies, the BS-ILS algorithm is compared with the MIRPPacs parallel algorithm from Munguia et al. (2019), executed under identical hardware conditions running Gentoo Linux 2.14 and 16 GB of RAM. The objective of this article is to demonstrate that it is possible to develop a MIRP solution method that is both effective and accessible,

without relying on commercial solvers or high-performance machines with dozens of threads, which are often unavailable. While the original MIRPPacs was executed on 96 threads, in this study, 16 threads were used, a configuration commonly available in modern PC processors. Consequently, all reported MIRPPacs results correspond to the 16-thread configuration, highlighting the practical applicability of the method under typical computational constraints. The results of BS-ILS and MIRPPacs are compared with the MIRPLib benchmarks.

The main results for instances with 120, 180, and 360 periods are summarized in Tables 5, 6, and 7, respectively. Complete results can be found in Appendices C and D. The tables include key metrics such as the value of N , the classification of each instance (Class), the objective values (Obj), the best percentage of the objective gap among the ten seeds (Best Gap), the average percentage of the objective gap (Average Gap) and the execution time (in hours) for BS-ILS and MIRPPacs methodologies. These tables show the best results obtained in 90,000 s (approximately 25 h). The reported results represent the best values obtained across all values tested for N . The proposed heuristic demonstrated improvements in the objective value of MIRPLib (Papageorgiou et al., 2018) in 18 different instances when $N = 1000$, considering the best seed of each instance. Furthermore, regarding the classification of instances, it can be observed that those classified as easy generally exhibit lower objective gaps compared to those classified as medium or hard. Also, for easy and medium instances increasing the number of nodes improves the results. However, this improvement is accompanied by a trade-off, specifically an increase in execution time. Regarding the MIRPPacs results, the execution times are worse than those of the BS-ILS heuristic. Although the best gap values are slightly better, the average results are very similar for the smaller instances; however, as the complexity of the instances increases, the performance declines considerably. This trend is illustrated more clearly and systematically in the boxplots. Figs. 8 and 9 present a comparison of BS-ILS in terms of the best gap and the average gap results, respectively, with the results separated by horizon size and also aggregated for general assessment. It can be observed that, especially for instances with a horizon of 360, the BS-ILS methodology outperforms MIRPPacs. This indicates that the proposed heuristic is more effective in handling larger and more complex instances, demonstrating both robustness and scalability relative to the MIRPPacs approach. Fig. 10 presents a comparison



(a) BestGap comparison separated by horizon.



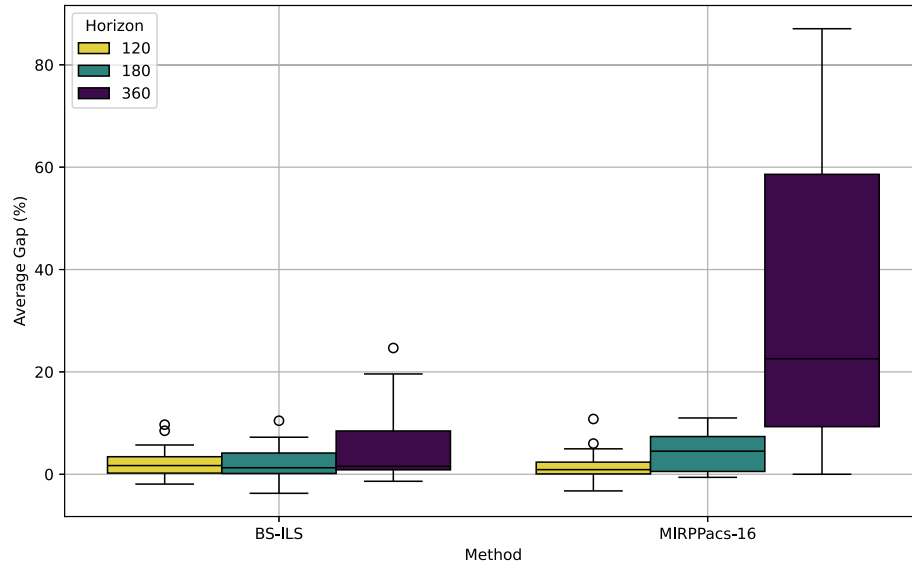
(b) BestGap comparison aggregating all horizons.

Fig. 8. Best gaps comparison for BS-ILS and MIRPPacs methods.

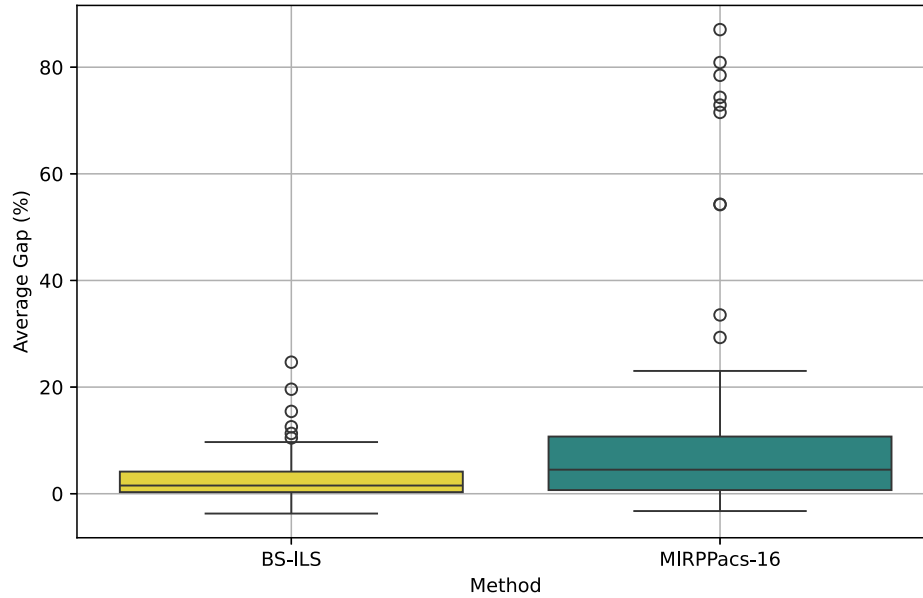
of execution times between the methodologies, highlighting that the heuristic significantly outperforms MIRPPacs in terms of efficiency, demonstrating substantially lower execution times across all tested instances. It can be observed that there is a large spread from the perspective of horizon segregation. This is due to the instances' difficulty levels (easy, medium, hard). Therefore, even the shorter-horizon instances can still be challenging in terms of execution time.

From the perspective of the easy, medium, and hard classification proposed by Papageorgiou et al. (2018), the boxplots presented in Figs. 11 and 12 illustrate the distribution of Gap Obj and time between different values of N and the classifications of instances in addition to MIRPPacs results. For Gap Obj, instances classified as easy consistently exhibit lower values, the median decreasing as N increases. These instances are relatively easier to solve, achieving near-optimal solutions

with a lower deviation. The medium class presents higher variability, especially for lower values of N , with a gradual reduction in dispersion as the size of the problem increases. The hard class maintains a stable Gap Obj distribution for $N = 10$ and $N = 100$. At $N = 1000$, the results worsen, as the algorithm cannot complete all stages within the available execution time. Analyzing MIRPPacs according to the class-based segregation, it can be observed that the performance varies across instance categories. For easy instances, MIRPPacs results are comparable to those of BS-ILS with $N = 1000$. For medium instances, the performance aligns more closely with BS-ILS at $N = 100$, while BS-ILS with $N = 1000$ clearly outperforms MIRPPacs. For hard instances, MIRPPacs consistently underperforms relative to all BS-ILS configurations, highlighting the heuristic's superior scalability and robustness across increasing instance complexity.



(a) Average Gap comparison by horizon.



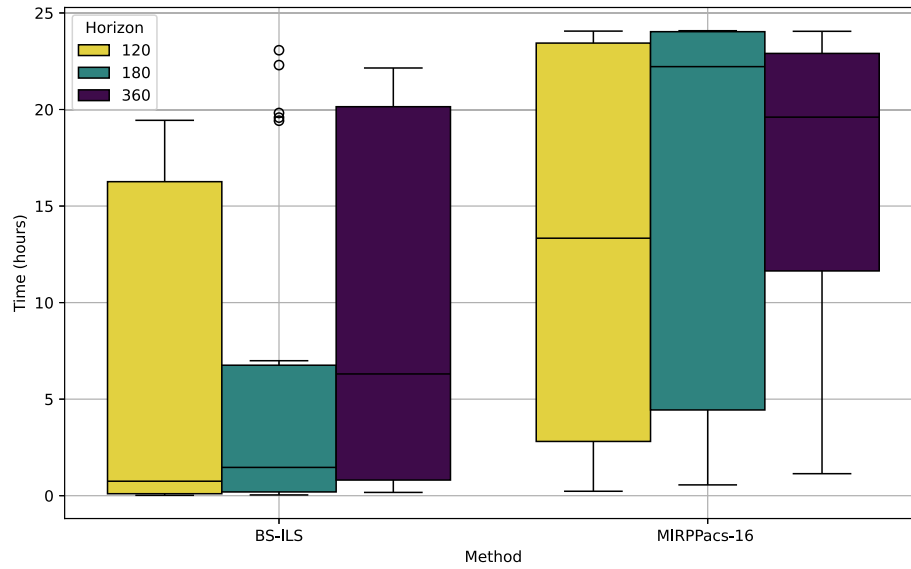
(b) Average Gap comparison aggregating all horizons.

Fig. 9. Average gaps comparison for BS-ILS and MIRPPacs methods.

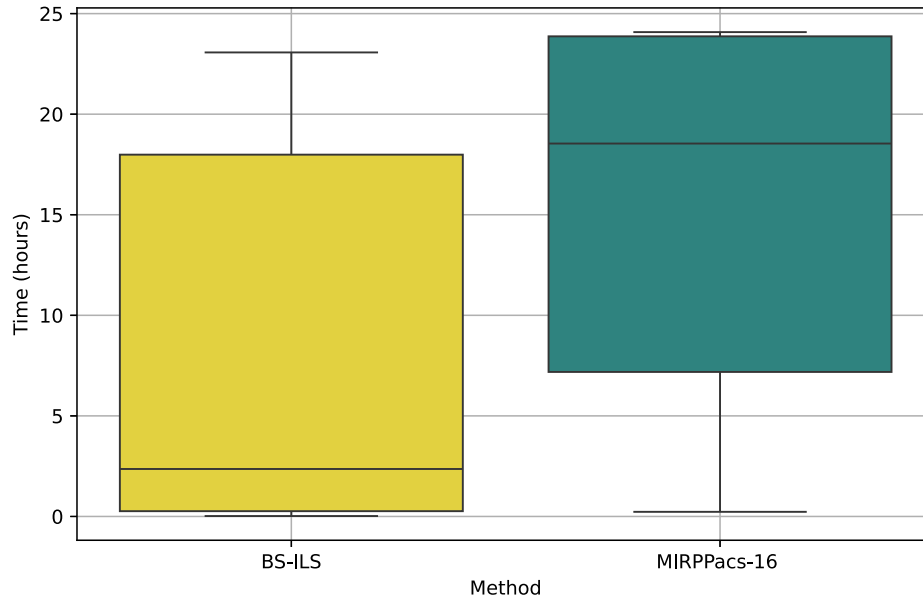
With respect to time, there is a clear increase in computational effort as N grows. For the easy class, execution times remain low in all values of N , indicating efficient convergence. In the medium class, execution times increase significantly, especially for higher N values ($N = 1000$), where a wider spread suggests inconsistent solver performance between different instances. The hard class consistently requires the highest computational effort, and as N grows, there is a substantial increase in execution time, confirming the difficulty of these instances. The execution time distribution in the hard class also exhibits larger variability, suggesting that some instances require substantially more computational resources. Across all scenarios, MIRPPacs results are substantially higher compared to those obtained with the BS-ILS methodology.

Our approach demonstrates a highly competitive performance when compared to MIRPPacs, under the same conditions. Overall, the method improves the best-known solutions for 18 instances, with an average best gap of -1.85% , and achieves competitive results on the remaining 54 instances, with an average best gap of 2.72% .

In terms of execution time, the proposed heuristic shows a clear advantage over MIRPPacs. The average execution time of our method is 7.69 h, whereas MIRPPacs requires 15.32 h, indicating that the benchmark takes approximately twice as long on average. This difference becomes even more evident when considering the median values. The median times are 2.36 h for the heuristic and 18.54 h for MIRPPacs. The substantial gap between these medians indicates that our approach is not only faster but also considerably more consistent, maintaining low execution times across the majority of instances. MIRPPacs



(a) Time comparison by horizon.



(b) Time comparison aggregating all horizons.

Fig. 10. Execution times comparison for BS-ILS and MIRPPacs methods.

presents greater variability and requires significantly longer processing times. Taken together, these findings highlight that the proposed domain-tailored heuristic can effectively balance solution quality and computational efficiency, making it a promising and practical tool for large and complex MIRPs and for real operational contexts that require fast and reliable decision making in maritime logistics planning.

5.3.2. BS and GRA results

As mentioned in Section 4.4, to effectively compare partial solutions, a greedy randomized algorithm is used to construct a complete solution starting from a given partial solution. To achieve this, different combinations of nondeterministic approaches for the GRA were tested: applying it separately to port selection, vessel selection, both or none.

It enables the algorithm to explore more diverse solution paths, increasing the likelihood of finding better solutions than the deterministic approach. For these tests, the 13 smallest Group 2 instances were used, with four different seeds, all with a 120-day horizon, and q was set to 5. The size of the instances allowed to set a large value for N , 1000. The configuration that achieved the best result, with the lowest aggregate average value among instances, was to apply randomness only to port selection while keeping the selection of the vessel deterministic. The second best was to apply randomness in both decisions followed by keeping greedy heuristic deterministic. The application of randomness only on vessel selection was the last. Table B.13 (Appendix B) presents the complete results of these tests.

Once it was established that nondeterministic methods would be used solely for the port, additional tests were performed to evaluate

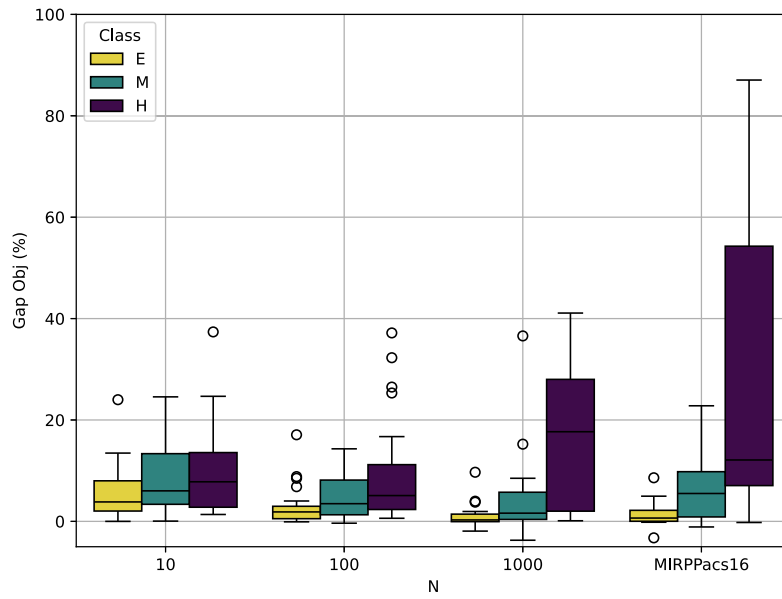


Fig. 11. Boxplot illustrating the objective gap (Gap Obj) across different N values (10, 100, 1000), comparing MIRPPacs and BS-ILS methodologies across instance classifications (E, M, H).

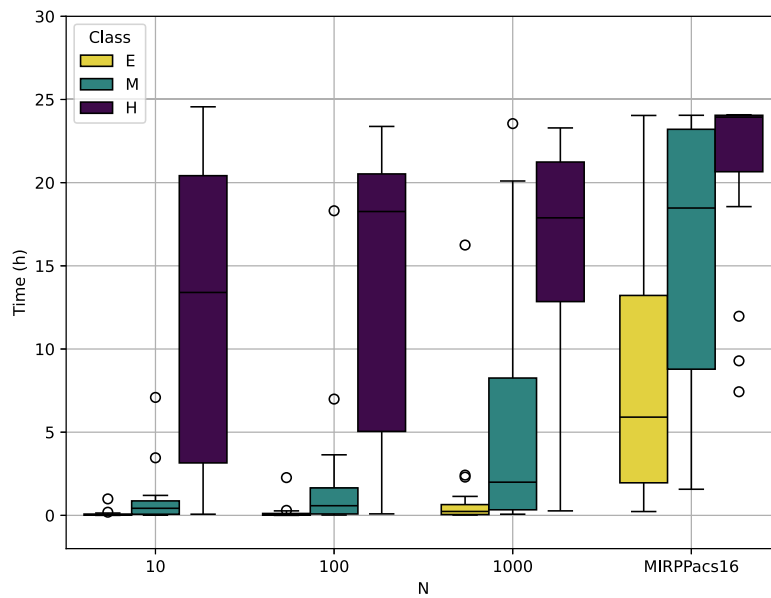


Fig. 12. Boxplot illustrating the execution time across different N values (10, 100, 1000), comparing MIRPPacs and BS-ILS methodologies across instance classifications (E, M, H).

three distinct techniques for aggregating the results of the GRA: minimum, mean, or median cost. The experiments were then conducted for the 13 smallest instances using three horizon sizes. For each instance, four executions with different seeds were carried out. The number of nodes allowed per level in the BS was set to 200. This decrease is attributed to the fact that instances now take into account a time horizon extending up to 360 days. The method that showed the best performance was the median with the lowest aggregate average value among the instances. The second best was the mean followed by the minimum. The complete results of these tests are shown in [Table B.14 B](#).

[Table 8](#) highlights the importance of the BS within the algorithm. Each column corresponds to a specific algorithm stage (After BS, After LS, and After ILS), allowing a comparison of the impact of BS on the results. The percentages indicate the improvement achieved by incorporating BS as opposed to relying solely on the GRA. Comparing the results obtained through the exclusive use of the GRA versus the application of the BS algorithm, it is possible to observe that the BS approach yields an average improvement of 18.40% in the solution following the BS stage. In particular, local search has a significant impact on results in the absence of BS, reducing the advantage of employing BS to 7.14%. When the complete heuristic is applied, the benefit of BS

Table 5

Summary of best and average results for each 120-period instance, including value of N , the classification of each instance (Class), the objective values (Obj), the best percentage of the objective gap (Best Gap %), the average percentage of the objective gap (Avg Gap %) and the execution time (in hours) for BS-ILS and MIRPPacs.

Instance	N	Class	Obj	BS-ILS Best gap (%)	BS-ILS Avg Gap (%)	BS-ILS Time (h)	MIRPPacs Best gap (%)	MIRPPacs Avg Gap (%)	MIRPPacs Time (h)
LR1_DR02_VC01_V6a	1000	E	33,809.00	0.00%	0.00%	0.02	0.00%	0.00%	0.23
LR1_DR02_VC02_V6a	1000	E	74,982.00	0.00%	0.00%	0.02	0.00%	1.73%	0.51
LR1_DR02_VC03_V7a	1000	E	40,446.00	-0.26%	-0.07%	0.03	-0.26%	4.17%	2.82
LR1_DR02_VC03_V8a	1000	E	43,721.00	0.00%	0.48%	0.02	0.00%	2.07%	1.78
LR1_DR02_VC04_V8a	1000	E	41,657.00	0.12%	0.30%	0.05	0.00%	0.46%	2.78
LR1_DR02_VC05_V8a	1000	E	36,659.00	-0.33%	-0.12%	0.05	-0.07%	0.60%	2.49
LR1_DR03_VC03_V10b	1000	M	92,941.00	1.21%	3.31%	0.12	3.10%	10.77%	1.57
LR1_DR03_VC03_V13b	1000	E	124,921.00	-0.10%	1.63%	0.30	0.03%	2.59%	10.12
LR1_DR03_VC03_V16a	1000	E	82,837.00	2.62%	9.69%	0.23	-4.69%	0.43%	7.33
LR1_DR04_VC03_V15a	1000	E	73,312.00	1.26%	1.95%	0.45	-0.38%	1.34%	6.75
LR1_DR04_VC03_V15b	1000	E	117,812.00	-4.27%	-1.92%	0.65	-5.18%	0.71%	13.28
LR1_DR04_VC05_V17a	1000	E	72,876.00	0.71%	1.30%	0.85	-0.35%	0.09%	7.49
LR1_DR04_VC05_V17b	1000	E	105,766.00	-3.84%	-1.61%	0.64	-6.32%	-3.25%	13.39
LR1_DR05_VC05_V25a	1000	E	105,328.00	0.07%	0.44%	2.30	-0.69%	-0.16%	15.76
LR1_DR05_VC05_V25b	1000	E	137,107.00	-0.89%	1.47%	2.42	-3.09%	2.09%	14.55
LR1_DR08_VC05_V38a	1000	M	166,615.00	1.28%	1.77%	14.72	-2.65%	-1.15%	23.44
LR1_DR08_VC05_V40a	1000	E	178,593.00	2.93%	3.79%	16.25	0.47%	4.97%	24.04
LR1_DR08_VC05_V40b	1000	M	200,746.00	6.13%	8.50%	19.44	0.52%	2.33%	22.80
LR1_DR08_VC10_V40a	1000	M	185,538.00	1.67%	2.32%	16.30	-0.47%	0.91%	23.44
LR1_DR08_VC10_V40b	1000	M	206,315.00	3.13%	5.63%	19.03	3.82%	6.01%	20.58
LR1_DR12_VC05_V70a	100	H	278,647.00	2.51%	3.88%	15.93	-1.75%	-0.22%	23.87
LR1_DR12_VC05_V70b	100	M	308,555.00	2.86%	5.72%	18.31	-0.43%	2.68%	24.05
LR1_DR12_VC10_V70a	100	H	283,154.00	2.34%	2.75%	17.26	-0.96%	0.91%	24.06
LR1_DR12_VC10_V70b	100	H	295,126.00	1.81%	2.34%	19.41	-1.67%	0.00%	24.06
Average				0.87%	2.23%	6.87	-0.88%	1.67%	12.97

Table 6

Summary of best and average results for each 180-period instance, including value of N , the classification of each instance (Class), the objective values (Obj), the best percentage of the objective gap (Best Gap %), the average percentage of the objective gap (Avg Gap %) and the execution time (in hours) for BS-ILS and MIRPPacs.

Instance	N	Class	Obj	BS-ILS Best Gap (%)	BS-ILS Avg Gap (%)	BS-ILS Time (h)	MIRPPacs Best Gap (%)	MIRPPacs Avg Gap (%)	MIRPPacs Time (h)
LR1_DR02_VC01_V6a	1000	E	52,167.00	0.00%	0.00%	0.04	0.00%	0.00%	0.56
LR1_DR02_VC02_V6a	1000	E	129,372.00	0.00%	0.28%	0.05	1.64%	2.21%	0.80
LR1_DR02_VC03_V7a	1000	M	60,547.00	0.00%	1.37%	0.07	0.09%	9.51%	4.59
LR1_DR02_VC03_V8a	1000	E	68,153.00	2.84%	3.98%	0.06	-1.00%	8.60%	3.40
LR1_DR02_VC04_V8a	1000	M	66,017.00	0.07%	0.39%	0.12	0.18%	0.81%	2.21
LR1_DR02_VC05_V8a	1000	M	58,619.00	-0.91%	-0.63%	0.09	-0.29%	0.63%	2.22
LR1_DR03_VC03_V10b	1000	M	125,638.00	-4.82%	-1.81%	0.22	-2.62%	5.11%	6.05
LR1_DR03_VC03_V13b	1000	E	165,764.00	-2.81%	-1.83%	0.55	-3.88%	-0.17%	13.04
LR1_DR03_VC03_V16a	1000	M	143,178.00	4.72%	7.24%	0.37	5.39%	10.96%	3.99
LR1_DR04_VC03_V15a	1000	M	118,621.00	0.21%	0.45%	1.06	-0.45%	0.35%	9.70
LR1_DR04_VC03_V15b	1000	E	189,989.00	-1.83%	-0.37%	1.14	-2.93%	0.29%	23.55
LR1_DR04_VC05_V17a	1000	M	117,710.00	0.70%	0.95%	1.64	-0.59%	-0.02%	16.12
LR1_DR04_VC05_V17b	1000	M	159,168.00	-4.74%	-3.72%	1.29	-4.16%	-0.65%	21.40
LR1_DR05_VC05_V25a	1000	M	171,620.00	-0.09%	0.21%	3.54	-0.95%	6.14%	23.87
LR1_DR05_VC05_V25b	1000	M	205,368.00	2.96%	6.84%	4.40	-0.58%	4.66%	23.96
LR1_DR08_VC05_V38a	1000	H	274,244.00	0.79%	1.18%	23.07	-1.53%	3.64%	23.81
LR1_DR08_VC05_V40a	100	H	296,760.00	1.53%	2.35%	5.19	-0.48%	3.88%	24.05
LR1_DR08_VC05_V40b	100	M	337,559.00	2.68%	4.66%	6.99	-1.50%	5.91%	24.03
LR1_DR08_VC10_V40a	100	H	304,261.00	1.17%	1.60%	6.68	-0.28%	4.32%	24.03
LR1_DR08_VC10_V40b	100	H	331,775.00	3.69%	6.19%	5.97	2.24%	7.08%	23.04
LR1_DR12_VC05_V70a	100	H	460,566.00	2.11%	2.63%	23.17	1.39%	8.41%	24.06
LR1_DR12_VC05_V70b	100	H	491,160.00	4.34%	7.26%	23.38	2.42%	10.12%	24.03
LR1_DR12_VC10_V70a	10	H	466,975.00	2.04%	2.77%	19.58	3.51%	7.98%	24.07
LR1_DR12_VC10_V70b	10	H	470,172.00	1.74%	2.85%	22.30	1.69%	7.17%	24.08
Average				0.68%	1.87%	6.29	-0.11%	4.46%	15.44

decreases further to 3.57% for this set of instances. However, its use remains beneficial, despite the increased computational effort. [Table B.15 B](#) shows the detailed results of the exclusive use of the GRA, without the BS.

Traditionally, the BS methodology does not involve solution completion for evaluation, relying instead on assessments of partial solutions. As already mentioned, a key novelty of this work is the development of a methodology to complete partial solutions and evaluate them

Table 7

Summary of best and average results for each 360-period instance, including value of N , the classification of each instance (Class), the objective values (Obj), the best percentage of the objective gap (Best Gap %), the average percentage of the objective gap (Avg Gap %) and the execution time (in hours) for BS-ILS and MIRPPacs.

Instance	N	Class	Obj	BS-ILS Best gap (%)	BS-ILS Avg Gap (%)	BS-ILS Time (h)	MIRPPacs Best gap (%)	MIRPPacs Avg Gap (%)	MIRPPacs Time (h)
LR1_DR02_VC01_V6a	1000	E	108,141.00	0.00%	0.00%	0.17	0.00%	0.00%	1.14
LR1_DR02_VC02_V6a	1000	E	283,031.00	-0.40%	0.29%	0.24	-0.42%	2.33%	5.06
LR1_DR02_VC03_V7a	1000	H	124,282.00	0.03%	1.63%	0.27	8.22%	23.03%	7.43
LR1_DR02_VC03_V8a	1000	M	141,166.00	0.91%	3.26%	0.23	7.26%	15.36%	10.56
LR1_DR02_VC04_V8a	1000	M	138,693.00	0.27%	0.37%	0.48	1.01%	6.33%	18.52
LR1_DR02_VC05_V8a	1000	H	122,598.00	-0.14%	0.12%	0.43	0.20%	10.05%	9.29
LR1_DR03_VC03_V10b	1000	H	259,888.00	-1.15%	0.96%	0.92	5.37%	14.10%	11.97
LR1_DR03_VC03_V13b	1000	M	316,441.00	-3.43%	-1.37%	2.01	1.07%	4.31%	18.43
LR1_DR03_VC03_V16a	1000	M	327,793.00	4.06%	6.10%	1.98	15.32%	22.75%	10.65
LR1_DR04_VC03_V15a	1000	M	252,710.00	0.41%	1.47%	4.63	0.29%	6.12%	17.37
LR1_DR04_VC03_V15b	1000	M	339,308.00	-1.90%	0.99%	4.29	6.74%	12.56%	22.58
LR1_DR04_VC05_V17a	1000	H	251,623.00	0.65%	0.96%	6.51	0.45%	7.05%	22.50
LR1_DR04_VC05_V17b	1000	M	304,507.00	-1.30%	2.18%	6.10	3.76%	10.67%	23.13
LR1_DR05_VC05_V25a	1000	H	368,628.00	0.30%	0.59%	17.88	12.15%	71.52%	20.49
LR1_DR05_VC05_V25b	1000	H	410,053.00	5.24%	8.30%	15.07	7.80%	22.36%	24.05
LR1_DR08_VC05_V38a	100	H	596,969.00	9.09%	15.44%	21.80	7.17%	29.31%	24.03
LR1_DR08_VC05_V40a	100	H	652,380.00	0.63%	1.08%	19.62	27.99%	72.90%	19.71
LR1_DR08_VC05_V40b	100	H	709,713.00	9.31%	11.31%	20.36	23.10%	33.53%	24.03
LR1_DR08_VC10_V40a	100	H	663,245.00	0.65%	1.25%	22.15	15.65%	80.88%	19.50
LR1_DR08_VC10_V40b	100	H	724,513.00	6.99%	8.94%	20.10	20.67%	74.35%	18.56
LR1_DR12_VC05_V70a	10	H	1,021,389.00	6.99%	12.56%	20.43	11.53%	54.30%	24.05
LR1_DR12_VC05_V70b	10	H	1,093,013.00	20.06%	24.67%	20.73	15.63%	54.22%	24.01
LR1_DR12_VC10_V70a	10	H	1,024,399.00	1.49%	2.10%	20.28	14.79%	78.48%	21.16
LR1_DR12_VC10_V70b	10	H	1,001,541.00	16.07%	19.60%	18.91	20.70%	87.05%	22.83
Average				3.11%	5.12%	10.23	9.44%	33.07%	17.54

Table 8

Percentage differences of the BS algorithm combined with the GRA compared to using the GRA alone. Results are shown for each instance and horizon, with objective values after BS, LS, and ILS.

Instance	Horizon	BS (%)	LS (%)	ILS (%)
LR1_DR02_VC01_V6a	180	0.00	0.00	0.00
LR1_DR02_VC01_V6a	360	-17.58	0.13	0.13
LR1_DR02_VC02_V6a	180	-4.92	-0.86	-0.86
LR1_DR02_VC02_V6a	360	-13.76	-4.72	-4.17
LR1_DR02_VC03_V7a	120	-15.72	-5.44	-5.37
LR1_DR02_VC03_V8a	120	-27.07	-8.52	-7.65
LR1_DR02_VC03_V8a	180	-25.14	-1.38	1.48
LR1_DR02_VC04_V8a	120	-9.63	-2.73	-0.54
LR1_DR02_VC05_V8a	120	-1.19	-1.58	-1.08
LR1_DR03_VC03_V13b	120	-13.27	-0.95	-0.86
LR1_DR03_VC03_V16a	120	-66.78	-53.68	-21.05
LR1_DR04_VC03_V15a	120	-22.78	-1.41	0.41
LR1_DR04_VC03_V15b	120	-23.91	-10.11	-5.12
LR1_DR04_VC05_V17a	120	-3.91	-0.73	-0.26
LR1_DR04_VC05_V17b	120	-30.25	-15.05	-8.64
Average		-18.40	-7.14	-3.57

to identify the best ones. However, comparing partial MIRP solutions can be very challenging. Different evaluation criteria were tested, but incomplete solutions resulted in ships and ports being scheduled at different times, making comparisons difficult and evaluations inaccurate. Consequently, all attempts to evaluate partial solutions without completing the total solution were found to be unsatisfactory. Partial solutions were therefore completed using the GRA, enabling meaningful comparisons within the BS framework, as shown in Table 9. For a balanced comparison, both the BS and the greedy heuristic were executed for equivalent execution time. All experiments were carried out with 15 Easy instances, $N = 10$ and 10 seeds. This approach, although computationally expensive, since it involves running a greedy algorithm for each node in the tree, demonstrated to be much more effective than other approaches for evaluating successors.

6. Conclusions

The main objective of this work was to develop a heuristic approach capable of solving a MIRP without relying on mathematical optimization models and therefore (commercial) MILP solvers, while also aiming to obtain high-quality solutions for MIRPLib Group 2 instances. To this end, we propose a method that combines BS with ILS. The proposed BS strategy allowed the exploration of a restricted subset of promising solution paths, significantly reducing computational effort while directing the search toward high-quality solutions. This was possible due to the implementation of a GRA, which played a crucial role in efficiently completing partial solutions during BS. Furthermore, (iterated) local search procedures demonstrated to be essential to refine and improve the quality of solutions, producing competitive results. In particular, a comparative analysis with MIRPLib benchmarks presented 18 new best-known solutions.

The proposed methodology is flexible enough to adjust its performance by modifying the number of nodes in the BS. This allows for a controlled trade-off between smaller gaps and higher execution times, or vice versa. For smaller instances, the methodology shows strong efficiency, reaching solutions with low optimality gaps within minutes, and in some cases surpassing the quality of known benchmark results. For larger instances, maintaining a low gap requires increasing execution time (through the number of nodes in the BS). Nevertheless, satisfactory results can still be achieved in less than 24 h.

Additionally, the BS-ILS algorithm provides a competitive solution for MIRP instances, as evidenced by its comparison with the MIRPPacs algorithm (Munguía et al., 2019) under identical hardware conditions. Unlike methods relying on commercial solvers or high-performance machines, BS-ILS achieves effective results on MIRPLib benchmarks, confirming that efficient MIRP solutions can be developed with limited computational resources.

Despite the promising results in terms of convergence and solution quality, certain limitations of the proposed method should be acknowledged. In some cases, the method may converge to less competitive solutions, indicating that additional diversification strategies

Table 9

BS performance comparison with and without integration of the GRA, expressed as percentage objective gaps. Results are presented by instance and horizon, showing gaps after BS, LS, and ILS stages.

Instance	Horizon	BS without GRA			BS with GRA		
		BS (%)	LS (%)	ILS (%)	BS (%)	LS (%)	ILS (%)
LR1_DR02_VC01_V6a	180	79.44	43.88	5.75	0.90	0.00	0.00
LR1_DR02_VC01_V6a	360	79.07	61.81	21.43	1.76	0.13	0.13
LR1_DR02_VC02_V6a	180	82.93	19.38	16.90	0.83	0.09	0.09
LR1_DR02_VC02_V6a	360	81.08	14.04	11.86	1.66	1.37	1.33
LR1_DR02_VC03_V7a	120	93.21	43.64	27.01	4.72	0.93	0.93
LR1_DR02_VC03_V8a	120	91.60	10.14	8.90	6.90	0.93	0.93
LR1_DR02_VC03_V8a	180	91.83	21.20	18.85	10.16	8.44	8.41
LR1_DR02_VC04_V8a	120	94.71	87.61	32.99	3.26	0.90	0.77
LR1_DR02_VC05_V8a	120	95.34	46.12	1.01	2.44	0.05	-0.12
LR1_DR03_VC08_V13b	120	87.05	80.08	50.23	6.36	2.87	1.87
LR1_DR03_VC03_V16a	120	91.57	53.33	31.41	19.25	17.32	14.01
LR1_DR04_VC03_V15a	120	93.53	79.51	15.35	7.06	3.28	2.52
LR1_DR04_VC03_V15b	120	91.18	81.30	39.54	4.90	0.46	0.07
LR1_DR04_VC05_V17a	120	94.68	85.80	16.06	5.57	2.02	1.42
LR1_DR04_VC05_V17b	120	93.47	85.14	25.71	6.21	1.76	1.34
Average		89.38	54.20	21.53	5.47	2.70	2.25

Table A.10

MIRP index and set.

Indices and sets	Definition
$t \in T$	Set of time periods
$vc \in VC$	Vessel classes
$j \in J^P (r \in R^P)$	Set of production (loading) ports (regions)
$j \in J^C (r \in R^C)$	Set of consumption (discharging) ports (regions)
$j \in J (r \in R)$	Set of all ports (regions): $J = J^P \cup J^C$ and $R = R^P \cup R^C$
$n \in \mathcal{N}$	Set of regular nodes or port-time pairs: $\mathcal{N} = \{n = (j, t) : j \in J, t \in T\}$
$n \in \mathcal{N}_{s,t}$	Set of all nodes (including a source node n_s and a sink node n_t)
A^{vc}	Set of arcs associated with vessel class $vc \in VC$
$A^{vc,inter}$	Set of inter-regional arcs associated with vessel class $vc \in VC$
$a \in F_n^{vc}$	Forward star (set of outgoing arcs) associated with node $n = (j, t) \in \mathcal{N}_{s,t}$ and vessel class $vc \in VC$
$a \in R_n^{vc}$	Reverse star (set of incoming arcs) associated with node $n = (j, t) \in \mathcal{N}_{s,t}$ and vessel class $vc \in VC$
$a \in F_n^{S^{vc,inter}}$	Forward star of inter-regional arcs associated with node $n = (j, t) \in \mathcal{N}_{s,t}$ and vessel class $vc \in VC$

Table A.11

MIRP parameters.

Parameter	Definition
$B_j(B_r)$	Number of berths (berth limit) at port $j \in J$ (in region $r \in R$)
C_a^{vc}	Cost for vessel in vessel class vc to traverse arc $a = ((j_1, t_1), (j_2, t_2)) \in A^{vc}$
$d_{j,t}$	Number of units that can be produced/consumed at port $j \in J$ in time period $t \in T$
$\Delta_j(\Delta_r)$	An indicator parameter taking value +1 if $j \in J^P (r \in R^P)$ and -1 if $j \in J^C (r \in R^C)$
$P_{j,t}$	Nonnegative penalty parameter associated with one unit of lost production or stockout at port j in time period t
Q^{vc}	Capacity of a vessel in vessel class vc
$S_{j,t}^{min} / S_{j,t}^{max}$	Lower/upper bound (capacity) at port $j \in J$ in time period $t \in T$
$s_{j,0}$	Initial inventory at port $j \in J$

could further increase its robustness. Also, as the problem scale increases, particularly in hard-instance classes, the variation of results also increases, showing that the method can be more sensitive in these situations. Moreover, performance tends to vary notably among different instance classes, with more complex ones generally requiring greater computational effort.

Future work includes the development of new local search operators and acceleration mechanisms to speed up the convergence of the

Table A.12

MIRP decision variables.

Decision variable	Definition
$\alpha_{j,t}$	(continuous) amount to produce/consume at port $j \in J$ in time period $t \in T$
$s_{j,t}$	(continuous) number of units of inventory at port $j \in J$ available at the end of period $t \in T$
x_a^{vc}	(integer) number of vessels in vessel class vc that take arc a

method. In addition, adaptive perturbation mechanisms could be designed to improve the diversification phase of the proposed algorithm, helping it escape local optima. Furthermore, integrating adaptive mechanisms aimed at improving convergence stability, scalability, and robustness across diverse operational scenarios would further strengthen the practical relevance of the proposed approach.

CRedit authorship contribution statement

Nathalie Sanghikian: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rafael Meirelles:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Anand Subramanian:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Rafael Martinelli:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

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Table B.13

Performance comparison of different types of randomness applied to the greedy heuristic: deterministic version (No Random), randomization over ports (Random Port), vessels (Random Vessel), and both (Both Random). Randomness prioritizes ports with earlier capacity violations and vessels with earlier arrival times, instead of selecting simply the first to violate constraints or serve ports.

Instances	No Random	Random Port	Random Vessel	Both Random
LR1_DR02_VC01_V6a	33,808.95	33,808.95	33,808.95	33,808.95
LR1_DR02_VC02_V6a	75,403.72	74,981.65	75,403.72	75,403.72
LR1_DR02_VC03_V7a	41,274.88	40,454.77	41,142.09	40,919.53
LR1_DR02_VC03_V8a	45,707.88	43,956.88	44,391.83	44,632.84
LR1_DR02_VC04_V8a	41,962.49	41,959.43	41,969.56	41,954.04
LR1_DR02_VC05_V8a	36,677.10	36,767.66	36,718.42	36,692.15
LR1_DR03_VC03_V10b	89,398.66	98,455.49	97,672.14	98,555.54
LR1_DR03_VC03_V11a	130,495.12	127,476.55	129,629.45	128,650.85
LR1_DR03_VC03_V11b	105,661.44	96,296.38	105,158.58	100,656.24
LR1_DR04_VC03_V15a	75,696.28	75,133.39	75,451.92	75,484.39
LR1_DR04_VC03_V15b	126,659.43	112,634.77	125,544.45	121,872.08
LR1_DR04_VC05_V17a	74,759.18	74,593.48	74,897.26	74,667.23
LR1_DR04_VC05_V17b	113,388.55	107,885.74	112,136.89	111,400.20
Average	76,222.59	74,185.01	76,455.79	75,745.98

Table B.14

Performance comparison of using different aggregation methods (minimum, mean, and median costs) for the GRA results.

Instance	Horizon	Minimum cost	Mean cost	Median cost
LR1_DR02_VC01_V6a	120	33,923.34	33,939.33	34,266.97
LR1_DR02_VC01_V6a	180	52,647.27	52,481.66	52,601.11
LR1_DR02_VC01_V6a	360	109,414.31	110,131.37	109,655.42
LR1_DR02_VC02_V6a	120	77,395.06	76,818.15	76,624.52
LR1_DR02_VC02_V6a	180	135,993.47	135,549.76	135,545.76
LR1_DR02_VC02_V6a	360	304,937.43	303,904.98	302,975.50
LR1_DR02_VC03_V7a	120	42,910.78	42,464.15	42,991.15
LR1_DR02_VC03_V7a	180	67,109.18	67,550.13	67,528.97
LR1_DR02_VC03_V7a	360	140,255.62	143,311.20	142,994.77
LR1_DR02_VC03_V8a	120	47,900.23	48,310.53	48,050.30
LR1_DR02_VC03_V8a	180	77,256.03	78,405.74	77,311.96
LR1_DR02_VC03_V8a	360	174,802.30	175,016.07	173,030.24
LR1_DR02_VC04_V8a	120	42,091.77	42,164.18	42,218.00
LR1_DR02_VC04_V8a	180	68,388.41	67,871.17	67,815.13
LR1_DR02_VC04_V8a	360	145,597.41	145,606.29	145,889.97
LR1_DR02_VC05_V8a	120	37,002.98	36,954.36	36,988.57
LR1_DR02_VC05_V8a	180	58,923.26	58,750.84	58,762.10
LR1_DR02_VC05_V8a	360	124,037.47	123,887.84	123,824.35
LR1_DR03_VC03_V10b	120	110,858.34	111,754.60	114,018.19
LR1_DR03_VC03_V10b	180	145,883.67	145,080.40	146,118.27
LR1_DR03_VC03_V10b	360	37,0964.74	362,337.30	358,205.90
LR1_DR03_VC03_V13b	120	136,033.99	134,909.53	135,552.79
LR1_DR03_VC03_V13b	180	173,509.93	172,567.81	172,606.80
LR1_DR03_VC03_V13b	360	341,677.79	341,456.29	341,661.21
LR1_DR03_VC03_V16a	120	138,716.00	137,684.78	131,597.83
LR1_DR03_VC03_V16a	180	276,705.77	276,930.69	274,921.66
LR1_DR03_VC03_V16a	360	682,519.75	682,040.67	682,703.66
LR1_DR04_VC03_V15a	120	82,961.41	85,314.85	84,632.05
LR1_DR04_VC03_V15a	180	148,329.62	148,596.19	148,713.50
LR1_DR04_VC03_V15a	360	301,765.71	301,804.37	301,325.94
LR1_DR04_VC03_V15b	120	143,300.31	143,387.59	143,254.13
LR1_DR04_VC03_V15b	180	212,916.42	212,891.02	212,375.42
LR1_DR04_VC03_V15b	360	387,058.14	382,095.79	381,325.83
LR1_DR04_VC05_V17a	120	75,673.42	75,783.27	75,735.27
LR1_DR04_VC05_V17a	180	122,075.49	121,668.81	122,024.03
LR1_DR04_VC05_V17a	360	261,484.75	260,646.48	260,256.08
LR1_DR04_VC05_V17b	120	134,376.88	133,850.50	132,773.17
LR1_DR04_VC05_V17b	180	197,886.09	198,980.83	199,055.59
LR1_DR04_VC05_V17b	360	412,700.92	408,949.66	409,822.29
Average		169,179.11	168,765.36	168,403.96

Appendix A. Maritime inventory routing problem model

This appendix presents the formulation that corresponds to the MIP model for “Group 2” instances available on MIRPLib (Papageorgiou, 2013). The “Group 2” instances describe long-horizon deterministic single product maritime inventory routing problems. In the following description, the term “port” and “region” are used as synonyms, since it is assumed that there is one port per region. Tables A.10–A.12 present

the indices and sets used in the model, the parameters, and the decision variables, respectively.

Appendix B. Detailed results of the greedy randomized algorithm

This appendix describes the tests to identify the best use of randomness in the GRA, shown in Table B.13. It also includes the experiments to identify the best aggregation strategy, summarized in Table B.14. In addition, Table B.15 presents the results of the GRA without the BS.

Table B.15

Results of the GRA without BS, showing instance, horizon, instance objective value, results after BS, LS, and ILS, along with the objective gap (%).

Instance	Horizon	Obj	After BS	After LS	After ILS	Gap obj
LR1_DR02_VC01_V6a	180	52,167.00	52,642.82	52,167.21	52,167.21	0.00%
	360	108,141.00	129,438.62	108,141.00	108,141.00	0.00%
LR1_DR02_VC02_V6a	180	129,372.00	136,876.18	130,603.20	130,603.20	0.95%
	360	283,031.00	327,394.47	300,509.56	298,807.87	5.57%
LR1_DR02_VC03_V7a	120	40,446.00	49,126.32	43,045.13	43,018.93	6.36%
LR1_DR02_VC03_V8a	120	43,721.00	59,676.14	47,892.73	47,508.72	8.66%
	180	68,153.00	94,931.65	75,464.21	73,314.49	7.57%
LR1_DR02_VC04_V8a	120	41,657.00	47,211.03	43,183.50	42,206.12	1.32%
LR1_DR02_VC05_V8a	120	36,659.00	38,024.22	37,255.92	37,008.85	0.95%
LR1_DR03_VC03_V13b	120	124,921.00	151,117.90	129,835.53	128,395.38	2.78%
LR1_DR03_VC03_V16a	120	82,837.00	171,093.48	153,981.07	116,603.27	40.76%
LR1_DR04_VC03_V15a	120	73,312.00	96,851.16	76,859.70	74,896.02	2.16%
LR1_DR04_VC03_V15b	120	117,812.00	153,494.96	130,322.79	123,931.88	5.19%
LR1_DR04_VC05_V17a	120	72,876.00	80,193.56	74,921.30	74,119.90	1.71%
LR1_DR04_VC05_V17b	120	105,766.00	146,881.76	123,865.09	116,463.71	10.11%

Table C.16

Results for instance with horizon = 120.

Instance	After BS		After LS		After ILS	
	Cost	Time (s)	Cost	Time (s)	Cost	Time (s)
LR1_DR02_VC01_V6a						
10	34,284.56	0.33	33,808.96	0.96	33,808.95	8.12
100	34,284.56	3.74	33,808.95	1.94	33,808.95	8.56
1000	34,284.55	46.87	33,808.95	11.58	33,808.95	9.20
LR1_DR02_VC02_V6a						
10	78,034.15	0.42	77,154.39	0.96	77,154.39	11.11
100	75,896.88	4.35	74,986.52	1.92	74,986.52	11.38
1000	75,896.88	55.58	74,981.65	12.71	74,981.65	12.50
LR1_DR02_VC03_V7a						
10	46,539.82	0.50	45,239.31	0.98	44,589.17	12.31
100	43,149.74	6.62	41,894.52	2.14	41,741.28	12.60
1000	41,181.83	79.40	40,422.70	14.64	40,418.33	13.09
LR1_DR02_VC03_V8a						
10	52,232.95	0.50	49,765.15	1.01	47,802.56	10.23
100	48,890.05	5.68	47,183.47	1.99	46,933.58	10.17
1000	45,087.94	57.36	43,933.34	12.84	43,933.34	10.44
LR1_DR02_VC04_V8a						
10	44,713.78	0.68	42,964.05	1.05	42,544.75	20.28
100	43,197.50	12.18	42,044.46	3.10	41,991.97	20.14
1000	42,898.48	118.92	41,869.48	24.52	41,781.14	21.21
LR1_DR02_VC05_V8a						
10	38,019.89	1.05	37,049.38	1.19	36,993.17	19.42
100	37,679.57	7.74	36,793.08	2.59	36,751.91	17.67
1000	37,568.00	123.63	36,618.41	23.39	36,615.31	21.11
LR1_DR03_VC03_V10b						
10	121,811.38	2.72	117,739.23	1.47	113,670.47	50.63
100	108,364.17	33.45	104,333.34	7.95	103,925.20	53.39
1000	99,476.72	323.64	96,408.11	68.16	96,121.92	54.90
LR1_DR03_VC03_V13b						
10	139,635.03	6.32	133,208.53	2.52	130,367.72	105.54
100	134,053.92	68.26	129,886.47	16.58	128,270.51	108.88
1000	130,717.41	840.38	127,200.36	144.65	126,991.77	110.73
LR1_DR03_VC03_V16a						
10	123,600.57	3.57	120,040.12	1.63	108,989.56	68.64
100	108,054.83	44.24	103,562.20	7.94	99,896.70	68.82
1000	94,693.57	667.04	92,361.45	75.19	91,722.54	75.64
LR1_DR04_VC03_V15a						
10	85,065.72	9.51	79,222.44	3.41	76,335.89	154.51
100	78,960.20	113.40	75,592.02	23.47	74,987.89	155.92
1000	77,869.10	1,200.35	75,071.05	243.11	74,766.27	161.78
LR1_DR04_VC03_V15b						
10	137,193.84	10.97	128,493.03	3.27	125,111.15	181.18
100	126,980.69	152.41	121,404.81	24.30	120,354.87	184.33
1000	120,744.97	1,922.73	115,816.45	241.11	115,595.16	189.55
LR1_DR04_VC05_V17a						
10	78,281.99	13.76	75,420.80	4.17	74,376.81	202.40
100	77,423.59	143.95	74,597.68	32.73	74,310.70	185.31
1000	76,615.59	2,501.47	74,140.90	347.03	73,838.28	192.97

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Table C.16 (continued).

LR1_DR04_VC05_V17b						
10	127,502.41	14.29	118,171.73	4.55	115,529.62	265.06
100	113,954.32	162.32	109,448.30	31.79	107,703.96	242.38
1000	109,238.23	1,768.79	104,654.66	306.22	104,088.07	243.10
LR1_DR05_VC05_V25a						
10	111,329.44	44.66	107,766.10	5.82	106,316.48	443.41
100	109,634.56	503.71	106,764.63	53.51	106,324.31	423.19
1000	108,797.89	7,291.29	106,045.50	550.34	105,788.46	425.97
LR1_DR05_VC05_V25b						
10	189,828.79	73.97	176,910.79	9.21	156,702.73	600.91
100	165,292.54	475.18	157,035.02	66.50	150,386.03	528.53
1000	147,300.61	7,311.00	142,548.89	782.86	139,153.80	610.03
LR1_DR08_VC05_V38a						
10	194,838.20	432.76	182,504.74	51.63	173,411.10	2,655.56
100	178,883.23	4,430.73	172,989.15	522.94	170,268.93	2,578.11
1000	175,908.41	44,682.83	170,705.34	5,494.52	169,611.77	2,806.44
LR1_DR08_VC05_V40a						
10	206,302.93	669.71	191,718.19	51.30	185,479.86	2,839.44
100	197,544.15	4,878.69	188,136.46	577.87	186,075.27	2,696.78
1000	193,497.63	50,505.37	187,247.40	5,425.69	185,635.34	2,564.87
LR1_DR08_VC05_V40b						
10	294,137.00	523.97	254,255.02	91.42	241,916.49	3,695.24
100	261,855.12	8,439.24	238,284.11	840.46	230,402.16	3,839.73
1000	241,820.77	64,755.20	221,305.01	6,988.89	219,405.49	3,538.98
LR1_DR08_VC10_V40a						
10	214,865.80	366.14	196,566.83	60.40	189,327.63	2,721.31
100	200,602.05	4,162.67	191,019.06	597.06	189,482.71	2,681.85
1000	196,626.96	53,120.24	190,814.21	6,190.46	189,947.48	2,777.48
LR1_DR08_VC10_V40b						
10	294,184.29	413.05	253,031.47	96.52	237,533.39	3,649.62
100	263,011.25	4,533.72	242,892.76	751.38	235,851.34	3,470.86
1000	238,482.36	58,405.56	220,928.41	6,707.73	218,618.51	3,393.75
LR1_DR12_VC05_V70a						
10	358,963.23	6,805.37	301,250.10	520.32	294,470.44	17,656.46
100	324,138.04	35,538.64	291,141.43	5,123.27	289,887.97	16,689.33
1000	353,403.69	69,669.61				
LR1_DR12_VC05_V70b						
10	419,761.70	5,578.49	347,070.28	513.35	332,374.61	19,430.86
100	367,113.06	41,974.45	331,918.15	4,591.31	327,280.22	19,359.10
1000	486,551.39	72,351.29				
LR1_DR12_VC10_V70a						
10	342,678.38	4,562.38	294,406.09	625.95	291,042.65	19,385.05
100	318,116.26	38,436.93	292,395.63	6,116.68	291,163.06	17,569.65
1000	323,744.31	48,003.14				
LR1_DR12_VC10_V70b						
10	404,643.40	3,662.38	315,753.11	698.42	302,979.08	24,047.65
100	362,549.46	41,152.61	304,299.47	6,118.43	302,191.66	22,619.75
1000	394,386.99	64,392.79				

Table C.17

Results for instance with horizon = 180.

Instance	After BS		After LS		After ILS	
	Cost	Time (s)	Cost	Time (s)	Cost	Time (s)
LR1_DR02_VC01_V6a						
10	52,720.93	0.61	52,167.21	1.10	52,167.21	22.74
100	52,642.81	6.62	52,167.21	2.65	52,167.21	21.08
1000	52,642.81	94.61	52,167.21	21.97	52,167.21	23.20
LR1_DR02_VC02_V6a						
10	135,607.07	0.97	134,441.25	1.12	134,108.80	33.54
100	131,539.56	10.18	130,564.85	3.52	130,557.38	33.96
1000	130,768.96	114.01	129,730.83	26.94	129,730.83	34.72
LR1_DR02_VC03_V7a						
10	69,543.39	1.16	67,806.47	1.27	67,403.21	37.76
100	66,039.27	15.02	64,314.43	4.34	64,254.44	38.97
1000	63,008.35	170.21	61,427.60	35.73	61,389.49	37.33
LR1_DR02_VC03_V8a						
10	83,146.02	1.23	79,509.29	1.20	78,766.63	31.08
100	75,867.13	13.82	74,669.41	3.62	74,480.52	28.99
1000	72,433.92	140.28	71,057.23	33.62	70,976.93	33.37
LR1_DR02_VC04_V8a						
10	70,509.01	2.21	69,140.70	1.47	68,527.69	64.20
100	67,633.81	29.72	66,554.79	7.03	66,321.93	60.82
1000	67,370.55	298.05	66,308.82	71.39	66,275.61	66.72

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Table C.17 (continued).

LR1_DR02_VC05_V8a						
10	59,608.12	1.52	58,753.09	1.38	58,651.06	56.72
100	59,282.38	23.91	58,479.19	5.68	58,408.43	58.32
1000	59,159.02	211.75	58,263.45	48.89	58,250.16	56.21
LR1_DR03_VC03_V10b						
10	151,547.46	3.69	144,812.72	2.86	141,354.97	143.09
100	133,851.75	53.78	129,640.70	17.22	128,894.80	138.44
1000	127,037.70	496.00	123,669.84	159.34	123,407.03	133.28
LR1_DR03_VC03_V13b						
10	177,884.14	9.47	170,719.50	5.91	166,326.54	282.12
100	170,313.54	123.41	166,347.12	39.97	165,608.02	275.62
1000	166,629.11	1,285.82	162,966.24	380.88	162,778.22	296.05
LR1_DR03_VC03_V16a						
10	205,182.19	10.16	196,349.58	3.83	188,838.47	234.60
100	169,685.17	87.86	165,355.27	23.95	162,513.73	220.53
1000	157,460.04	935.01	154,533.69	187.96	154,345.81	209.90
LR1_DR04_VC03_V15a						
10	136,629.78	19.05	128,501.79	8.03	122,834.28	442.86
100	125,081.25	210.23	121,771.57	62.76	120,301.85	429.05
1000	122,580.25	2,759.45	119,374.31	628.76	119,162.69	438.06
LR1_DR04_VC03_V15b						
10	208,480.24	20.68	197,733.52	8.80	194,185.70	483.37
100	199,172.53	324.28	192,646.92	71.73	191,604.05	466.71
1000	194,556.34	3,014.38	189,761.19	621.67	189,297.78	478.37
LR1_DR04_VC05_V17a						
10	124,131.41	47.51	120,540.91	9.10	119,450.53	634.71
100	122,376.71	291.29	119,737.27	82.81	119,169.55	601.81
1000	121,662.90	4,473.17	119,033.68	843.88	118,838.48	570.71
LR1_DR04_VC05_V17b						
10	180,520.24	28.78	171,924.08	11.43	167,156.45	744.99
100	167,678.48	306.95	161,045.86	92.94	159,886.94	714.37
1000	158,940.40	3,208.26	154,259.86	754.12	153,453.60	674.34
LR1_DR05_VC05_V25a						
10	182,232.86	87.61	175,859.22	12.04	172,820.56	1,434.46
100	178,172.94	1,115.46	173,171.72	143.24	172,325.42	1,475.40
1000	175,032.22	9,797.75	172,148.67	1,483.70	171,987.29	1,452.28
LR1_DR05_VC05_V25b						
10	308,504.56	91.56	283,435.54	28.92	257,542.54	1,814.44
100	264,347.07	1,009.71	246,944.85	314.67	239,671.16	1,739.80
1000	232,105.24	11,557.28	221,614.12	2,560.21	220,445.42	1,704.34
LR1_DR08_VC05_V38a						
10	371,480.24	595.54	314,541.76	182.64	301,079.85	8,950.27
100	298,749.85	6,470.44	288,802.34	1,495.88	285,900.35	8,643.06
1000	282,815.20	82,539.17	277,523.54	83,056.87		
LR1_DR08_VC05_V40a						
10	362,875.57	952.43	334,655.98	191.92	318,086.76	9,941.73
100	321,263.83	7,298.19	309,711.46	1,785.72	303,891.24	9,583.60
1000	309,388.12	83,843.73	304,473.57	83,843.73		
LR1_DR08_VC05_V40b						
10	448,383.20	680.67	388,144.10	269.59	371,294.77	11,514.73
100	401,221.21	11,926.89	360,592.84	2,443.36	354,063.17	10,782.68
1000	398,217.73	84,791.16				
LR1_DR08_VC10_V40a						
10	359,409.15	712.97	331,860.61	191.92	314,525.30	9,805.86
100	322,553.82	12,322.27	311,011.07	1,871.89	309,214.55	9,868.49
1000	316,992.33	83,769.43	309,960.18	83,769.43		
LR1_DR08_VC10_V40b						
10	458,609.08	977.95	390,319.45	273.71	379,197.06	11,718.67
100	404,455.77	7,777.14	360,180.60	2,601.81	353,654.23	11,098.52
1000	436,670.63	80,973.98	364,424.03	86,060.43		
LR1_DR12_VC05_V70a						
10	579,863.38	9,532.19	503,449.29	1,696.28	490,463.48	58,690.48
100	510,997.33	78,033.03	472,986.02	83,414.40		
1000	614,880.49	67,167.73				
LR1_DR12_VC05_V70b						
10	685,043.40	11,057.53	569,804.84	1,699.07	548,540.09	62,655.70
100	594,658.83	83,354.89	529,627.05	841.73		
1000	758,276.83	56,830.38				
LR1_DR12_VC10_V70a						
10	551,861.95	9,160.68	493,103.31	1,853.61	480,276.39	59,463.88
100	516,280.02	80,404.01	482,322.51	82,055.62		
1000	530,070.26	78,033.50				
LR1_DR12_VC10_V70b						
10	671,368.54	9,618.47	519,846.82	2,033.55	483,959.57	71,001.10
100	629,621.29	74,072.13				
1000	655,532.88	63,163.95				

Table C.18
Results for instance with horizon = 360.

Instance	After BS		After LS		After ILS	
	Cost	Time (s)	Cost	Time (s)	Cost	Time (s)
LR1_DR02_VC01_V6a						
10	130,539.41	2.44	118,266.77	2.48	115,843.86	150.92
100	110,892.68	31.46	108,620.51	13.62	108,620.51	155.58
1000	109,040.08	317.54	108,141.00	124.08	108,141.00	156.50
LR1_DR02_VC02_V6a						
10	304,515.50	3.18	300,734.67	3.16	299,521.68	243.92
100	292,191.03	51.23	290,888.53	19.22	290,580.88	246.23
1000	284,439.15	448.87	283,870.25	148.21	283,857.52	261.71
LR1_DR02_VC03_V7a						
10	152,042.38	3.86	143,757.90	2.80	141,106.36	262.62
100	133,949.68	45.57	132,484.79	21.92	132,362.41	243.74
1000	127,797.24	519.52	126,410.03	184.77	126,345.86	257.35
LR1_DR02_VC03_V8a						
10	187,266.72	3.77	174,290.07	3.43	164,152.23	205.57
100	155,947.99	38.97	152,180.74	16.82	151,800.49	188.41
1000	147,778.88	485.08	145,925.59	146.21	145,925.59	203.97
LR1_DR02_VC04_V8a						
10	148,198.39	5.97	144,417.19	3.96	143,241.23	434.92
100	142,551.02	86.25	140,810.64	37.06	140,564.63	449.05
1000	140,277.82	929.64	139,203.20	349.89	139,203.18	445.34
LR1_DR02_VC05_V8a						
10	126,489.23	5.97	125,196.50	3.57	124,416.00	366.61
100	124,555.52	64.55	123,483.14	28.41	123,355.04	379.68
1000	123,631.38	870.44	122,787.01	286.55	122,751.13	391.93
LR1_DR03_VC03_V10b						
10	329,005.29	19.19	316,492.45	12.95	311,526.16	784.46
100	287,770.76	150.14	284,352.28	76.14	283,648.62	718.62
1000	264,787.09	1,989.88	262,465.56	621.96	262,404.37	715.45
LR1_DR03_VC03_V13b						
10	353,058.92	38.93	334,073.00	27.74	326,311.31	1,431.40
100	323,487.65	352.31	319,843.96	212.87	319,271.20	1,496.88
1000	315,786.35	3,882.89	312,402.74	1,841.60	312,157.14	1,496.86
LR1_DR03_VC03_V16a						
10	464,005.55	32.09	448,304.05	36.50	434,508.67	1,907.90
100	384,023.34	316.64	379,310.66	193.97	377,181.50	1,599.60
1000	354,108.05	3,576.99	349,629.10	1,847.91	349,105.55	1,701.45
LR1_DR04_VC03_V15a						
10	306,255.96	84.77	282,387.48	44.42	265,637.51	2,965.83
100	276,627.95	1,200.06	265,311.15	592.50	261,878.95	3,292.68
1000	262,160.28	9,398.41	257,619.15	4,210.93	256,489.27	3,040.24
LR1_DR04_VC03_V15b						
10	409,808.79	91.82	371,919.94	59.10	355,247.88	2,858.65
100	368,750.85	711.51	354,881.19	490.75	351,541.55	2,738.45
1000	348,977.76	9,089.58	343,490.65	3,542.88	342,712.78	2,802.89
LR1_DR04_VC05_V17a						
10	265,462.77	120.91	257,788.34	50.13	255,095.83	4,630.41
100	258,852.14	1,284.91	255,337.92	606.57	254,773.64	4,923.68
1000	256,800.04	13,461.00	254,228.58	5,447.35	254,070.18	4,543.56
LR1_DR04_VC05_V17b						
10	383,495.49	137.88	359,229.17	62.52	343,474.07	4,015.15
100	339,930.53	1,023.46	330,339.58	441.84	328,622.55	3,968.80
1000	319,466.58	13,662.85	311,613.92	4,161.99	311,308.52	4,128.22
LR1_DR05_VC05_V25a						
10	403,267.91	267.29	388,141.19	65.26	374,062.02	11,806.42
100	396,205.07	2,880.46	378,577.94	829.11	372,757.55	11,634.78
1000	391,689.44	34,648.71	371,419.79	19,606.67	370,808.16	13,003.51
LR1_DR05_VC05_V25b						
10	597,019.83	324.32	569,855.01	93.74	517,983.14	11,979.11
100	558,659.20	3,764.06	505,231.32	2,221.20	492,407.02	12,022.96
1000	483,005.51	27,864.72	450,911.35	15,412.65	447,158.01	10,981.34
LR1_DR08_VC05_V38a						
10	1,085,452.85	3,597.14	1,056,506.28	703.80	953,116.12	76,851.11
100	842,947.79	29,114.28	718,768.17	29,981.51	705,961.38	19,399.75
1000	1,013,341.61	77,596.52				
LR1_DR08_VC05_V40a						
10	893,239.46	3,619.93	747,650.70	3,165.75	718,559.97	81,619.35
100	691,576.42	28,895.76	661,190.48	13,159.22	659,529.96	28,563.72
1000	896,185.52	69,519.59				
LR1_DR08_VC05_V40b						
10	1,027,532.87	4,386.66	894,684.28	2,544.06	860,569.40	66,478.63
100	866,911.69	34,915.93	804,352.30	17,673.98	800,182.33	20,688.84
1000	1,058,391.25	73,126.73				

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Table C.18 (continued).

LR1_DR08_VC10_V40a						
10	838,314.16	2,888.32	716,248.93	2,530.10	698,379.30	81,925.74
100	703,775.17	34,944.33	672,834.22	13,742.11	671,630.12	31,041.66
1000	864,507.81	79,552.23				
LR1_DR08_VC10_V40b						
10	838,314.16	2,888.32	716,248.93	2,530.10	841,613.19	79,051.32
100	872,595.42	30,387.71	800,821.96	17,211.06	795,645.23	24,773.33
1000	1,051,389.35	71,734.38				
LR1_DR12_VC05_V70a						
10	1,498,705.74	26,035.35	1,209,606.76	14,909.76	1,168,105.20	73,556.17
100	1,508,194.51	69,379.01				
1000	1,610,962.69	38,255.60				
LR1_DR12_VC05_V70b						
10	1,734,036.19	28,433.57	1,483,546.39	18,342.76	1,450,879.45	74,633.02
100	1,739,513.17	76,371.96				
1000	1,786,227.29	55,203.71				
LR1_DR12_VC10_V70a						
10	1,154,238.35	32,098.33	1,049,776.20	11,667.49	1,046,321.06	73,001.60
100	1,148,804.41	72,442.23				
1000	1,194,132.95	45,688.29				
LR1_DR12_VC10_V70b						
10	1,393,107.77	28,548.83	1,257,955.30	6,542.26	1,245,728.22	68,081.75
100	1,362,528.96	37,393.17				
1000	1,375,430.96	45,631.04				

Table C.19

Summary of best and average results for each 120-period instance, including value of N , the classification of each instance (Class), the objective values (Obj), the best total cost among the ten runs (Best Cost), the best percentage of the objective gap (Best Gap %), the average total costs (Avg Cost), the average percentage of the objective gap (Avg Gap %) and the execution time (in hours).

Instance	N	Class	Obj	Best Cost	Best Gap (%)	Avg Cost	Avg Gap (%)	Time (h)
LR1_DR02_VC01_V6a	1000	E	33,809.00	33,808.95	0.00%	33,808.95	0.00%	0.02
LR1_DR02_VC02_V6a	1000	E	74,982.00	74,981.65	0.00%	74,981.65	0.00%	0.02
LR1_DR02_VC03_V7a	1000	E	40,446.00	40,340.01	-0.26%	40,418.33	-0.07%	0.03
LR1_DR02_VC03_V8a	1000	E	43,721.00	43,721.43	0.00%	43,933.34	0.48%	0.02
LR1_DR02_VC04_V8a	1000	E	41,657.00	41,708.65	0.12%	41,781.14	0.30%	0.05
LR1_DR02_VC05_V8a	1000	E	36,659.00	36,536.62	-0.33%	36,615.31	-0.12%	0.05
LR1_DR03_VC03_V10b	1000	M	92,941.00	94,077.04	1.21%	96,121.92	3.31%	0.12
LR1_DR03_VC03_V13b	1000	E	124,921.00	124,793.68	-0.10%	126,991.77	1.63%	0.30
LR1_DR03_VC03_V16a	1000	E	82,837.00	85,065.66	2.62%	91,722.54	9.69%	0.23
LR1_DR03_VC03_V15a	1000	E	73,312.00	74,250.27	1.26%	74,766.27	1.95%	0.45
LR1_DR04_VC03_V15b	1000	E	117,812.00	112,990.44	-4.27%	115,595.16	-1.92%	0.65
LR1_DR04_VC05_V17a	1000	E	72,876.00	73,394.78	0.71%	73,838.28	1.30%	0.85
LR1_DR04_VC05_V17b	1000	E	105,766.00	101,855.06	-3.84%	104,088.07	-1.61%	0.64
LR1_DR05_VC05_V25a	1000	E	105,328.00	105,396.95	0.07%	105,788.46	0.44%	2.30
LR1_DR05_VC05_V25b	1000	E	137,107.00	135,896.32	-0.89%	139,153.80	1.47%	2.42
LR1_DR08_VC05_V38a	1000	M	166,615.00	168,770.52	1.28%	169,611.77	1.77%	14.72
LR1_DR08_VC05_V40a	1000	E	178,593.00	183,980.41	2.93%	185,635.34	3.79%	16.25
LR1_DR08_VC05_V40b	1000	M	200,746.00	213,861.83	6.13%	219,405.49	8.50%	19.44
LR1_DR08_VC10_V40a	1000	M	185,538.00	188,695.26	1.67%	189,947.48	2.32%	16.30
LR1_DR08_VC10_V40b	1000	M	206,315.00	212,975.63	3.13%	218,618.51	5.63%	19.03
LR1_DR12_VC05_V70a	100	H	278,647.00	285,822.72	2.51%	289,887.97	3.88%	15.93
LR1_DR12_VC05_V70b	100	M	308,555.00	317,650.75	2.86%	327,280.22	5.72%	18.31
LR1_DR12_VC10_V70a	100	H	283,154.00	289,926.06	2.34%	291,163.06	2.75%	17.26
LR1_DR12_VC10_V70b	100	H	295,126.00	300,566.09	1.81%	302,191.66	2.34%	19.41

Table C.20

Summary of best and average results for each 180-period instance, including value of N , the classification of each instance (Class), the objective values (Obj), the best total cost among the ten runs (Best Cost), the best percentage of the objective gap (Best Gap %), the average total costs (Avg Cost), the average percentage of the objective gap (Avg Gap %), and the execution time (in hours).

Instance	N	Class	Obj	Best Cost	Best Gap (%)	Avg Cost	Avg Gap (%)	Time (h)
LR1_DR02_VC01_V6a	1000	E	52,167.00	52,167.21	0.00%	52,167.21	0.00%	0.04
LR1_DR02_VC02_V6a	1000	E	129,372.00	129,372.06	0.00%	129,730.83	0.28%	0.05
LR1_DR02_VC03_V7a	1000	M	60,547.00	60,546.80	0.00%	61,389.49	1.37%	0.07
LR1_DR02_VC03_V8a	1000	E	68,153.00	70,143.11	2.84%	70,976.93	3.98%	0.06
LR1_DR02_VC04_V8a	1000	M	66,017.00	66,064.00	0.07%	66,275.61	0.39%	0.12
LR1_DR02_VC05_V8a	1000	M	58,619.00	58,090.14	-0.91%	58,250.16	-0.63%	0.09
LR1_DR03_VC03_V10b	1000	M	125,638.00	119,855.91	-4.82%	123,407.03	-1.81%	0.22
LR1_DR03_VC03_V13b	1000	E	165,764.00	161,232.16	-2.81%	162,778.22	-1.83%	0.55
LR1_DR03_VC03_V16a	1000	M	143,178.00	150,277.58	4.72%	154,345.81	7.24%	0.37

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Table C.20 (continued).

Instance	N	Class	Obj	Best Cost	Best Gap (%)	Avg Cost	Avg Gap (%)	Time (h)
LR1_DR04_VC03_V15a	1000	M	118,621.00	118,867.19	0.21%	119,162.69	0.45%	1.06
LR1_DR04_VC03_V15b	1000	E	189,989.00	186,581.80	-1.83%	189,297.78	-0.37%	1.14
LR1_DR04_VC05_V17a	1000	M	117,710.00	118,538.31	0.70%	118,838.48	0.95%	1.64
LR1_DR04_VC05_V17b	1000	M	159,168.00	151,960.44	-4.74%	153,453.60	-3.72%	1.29
LR1_DR05_VC05_V25a	1000	M	171,620.00	171,468.45	-0.09%	171,987.29	0.21%	3.54
LR1_DR05_VC05_V25b	1000	M	205,368.00	211,634.13	2.96%	220,445.42	6.84%	4.40
LR1_DR08_VC05_V38a	1000	H	274,244.00	276,414.18	0.79%	277,523.54	1.18%	23.07
LR1_DR08_VC05_V40a	100	H	296,760.00	301,376.70	1.53%	303,891.24	2.35%	5.19
LR1_DR08_VC05_V40b	100	M	337,559.00	346,867.62	2.68%	354,063.17	4.66%	6.99
LR1_DR08_VC10_V40a	100	H	304,261.00	307,870.53	1.17%	309,214.55	1.60%	6.68
LR1_DR08_VC10_V40b	100	H	331,775.00	344,475.68	3.69%	353,654.23	6.19%	5.97
LR1_DR12_VC05_V70a	100	H	460,566.00	470,486.52	2.11%	472,986.02	2.63%	23.17
LR1_DR12_VC05_V70b	100	H	491,160.00	513,439.11	4.34%	529,627.05	7.26%	23.38
LR1_DR12_VC10_V70a	10	H	466,975.00	476,715.45	2.04%	480,276.39	2.77%	19.58
LR1_DR12_VC10_V70b	10	H	470,172.00	478,507.98	1.74%	483,959.57	2.85%	22.30

Table C.21

Summary of best and average results for each 360-period instance, including value of N , the classification of each instance (Class), the objective values (Obj), the best total cost among the ten runs (Best Cost), the best percentage of the objective gap (Best Gap %), the average total costs (Avg Cost), the average percentage of the objective gap (Avg Gap %) and the execution time (in hours).

Instance	N	Class	Obj	Best Cost	Best Gap (%)	Avg Cost	Avg Gap (%)	Time (h)
LR1_DR02_VC01_V6a	1000	E	108,141.00	108,141.00	0.00%	108,141.00	0.00%	0.17
LR1_DR02_VC02_V6a	1000	E	283,031.00	281,910.11	-0.40%	283,857.52	0.29%	0.24
LR1_DR02_VC03_V7a	1000	H	124,282.00	124,315.00	0.03%	126,345.86	1.63%	0.27
LR1_DR02_VC03_V8a	1000	M	141,166.00	142,461.48	0.91%	145,925.59	3.26%	0.23
LR1_DR02_VC04_V8a	1000	M	138,693.00	139,072.01	0.27%	139,203.18	0.37%	0.48
LR1_DR02_VC05_V8a	1000	H	122,598.00	122,430.70	-0.14%	122,751.13	0.12%	0.43
LR1_DR03_VC03_V10b	1000	H	259,888.00	256,927.82	-1.15%	262,404.37	0.96%	0.92
LR1_DR03_VC03_V13b	1000	M	316,441.00	305,950.39	-3.43%	312,157.14	-1.37%	2.01
LR1_DR03_VC03_V16a	1000	M	327,793.00	341,662.77	4.06%	349,105.55	6.10%	1.98
LR1_DR04_VC03_V15a	1000	M	252,710.00	253,743.29	0.41%	256,489.27	1.47%	4.63
LR1_DR04_VC03_V15b	1000	M	339,308.00	332,981.91	-1.90%	342,712.78	0.99%	4.29
LR1_DR04_VC05_V17a	1000	H	251,623.00	253,260.87	0.65%	254,070.18	0.96%	6.51
LR1_DR04_VC05_V17b	1000	M	304,507.00	300,607.25	-1.30%	311,308.52	2.18%	6.10
LR1_DR05_VC05_V25a	1000	H	368,628.00	369,727.34	0.30%	370,808.16	0.59%	17.88
LR1_DR05_VC05_V25b	1000	H	410,053.00	432,720.23	5.24%	447,158.01	8.30%	15.07
LR1_DR08_VC05_V38a	100	H	596,969.00	656,654.32	9.09%	705,961.38	15.44%	21.80
LR1_DR08_VC05_V40a	100	H	652,380.00	656,498.46	0.63%	659,529.96	1.08%	19.62
LR1_DR08_VC05_V40b	100	H	709,713.00	782,566.17	9.31%	800,182.33	11.31%	20.36
LR1_DR08_VC10_V40a	100	H	663,245.00	667,557.67	0.65%	671,630.12	1.25%	22.15
LR1_DR08_VC10_V40b	100	H	724,513.00	777,250.17	6.99%	795,645.23	8.94%	20.10
LR1_DR12_VC05_V70a	10	H	1,021,389.00	1,098,138.01	6.99%	1,168,105.20	12.56%	20.43
LR1_DR12_VC05_V70b	10	H	1,093,013.00	1,367,218.87	20.06%	1,450,879.48	24.67%	20.73
LR1_DR12_VC10_V70a	10	H	1,024,399.00	1,039,843.49	1.49%	1,046,321.06	2.10%	20.28
LR1_DR12_VC10_V70b	10	H	1,001,541.00	1,193,328.37	16.07%	1,245,728.22	19.60%	18.91

Table D.22

MIRPPacs results for instance with horizon = 120.

Instance	Horizon	Class	Obj	Best Cost	Best GAP (%)	Avg Cost	Avg GAP (%)	Time (s)	Time (h)
LR1_DR02_VC01_V6a	120	E	33,809.00	33,808.95	0.00%	33,808.95	0.00%	824.40	0.23
LR1_DR02_VC02_V6a	120	E	74,982.00	74,981.66	0.00%	76,305.15	1.73%	1,837.43	0.51
LR1_DR02_VC03_V7a	120	E	40,446.00	40,340.01	-0.26%	42,204.16	4.17%	10,136.83	2.82
LR1_DR02_VC03_V8a	120	E	43,721.00	43,721.43	0.00%	44,643.11	2.07%	6,402.66	1.78
LR1_DR02_VC04_V8a	120	E	41,657.00	41,657.49	0.00%	41,847.54	0.46%	9,993.91	2.78
LR1_DR02_VC05_V8a	120	E	36,659.00	36,633.77	-0.07%	36,880.33	0.60%	8,950.80	2.49
LR1_DR03_VC03_V10b	120	M	92,941.00	95,910.43	3.10%	104,155.99	10.77%	5,638.36	1.57
LR1_DR03_VC03_V13b	120	E	124,921.00	124,952.24	0.03%	128,238.44	2.59%	36,435.90	10.12
LR1_DR03_VC03_V16a	120	E	82,837.00	79,128.42	-4.69%	83,194.45	0.43%	26,400.48	7.33
LR1_DR04_VC03_V15a	120	E	73,312.00	73,034.95	-0.38%	74,309.69	1.34%	24,302.70	6.75
LR1_DR04_VC03_V15b	120	E	117,812.00	112,007.12	-5.18%	118,657.45	0.71%	47,797.76	13.28
LR1_DR04_VC05_V17a	120	E	72,876.00	72,622.82	-0.35%	72,938.40	0.09%	26,959.34	7.49
LR1_DR04_VC05_V17b	120	E	105,766.00	99,478.44	-6.32%	102,438.73	-3.25%	48,197.65	13.39
LR1_DR05_VC05_V25a	120	E	105,328.00	104,607.92	-0.69%	105,158.45	-0.16%	56,732.38	15.76
LR1_DR05_VC05_V25b	120	E	137,107.00	132,994.49	-3.09%	140,036.81	2.09%	52,365.42	14.55
LR1_DR08_VC05_V38a	120	M	166,615.00	162,312.30	-2.65%	164,721.92	-1.15%	84,389.95	23.44
LR1_DR08_VC05_V40a	120	E	178,593.00	179,439.09	0.47%	187,929.39	4.97%	86,558.41	24.04
LR1_DR08_VC05_V40b	120	M	200,746.00	201,790.82	0.52%	205,526.43	2.33%	82,077.90	22.80
LR1_DR08_VC10_V40a	120	M	185,538.00	184,663.04	-0.47%	187,238.42	0.91%	84,376.94	23.44

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Table D.22 (continued).

Instance	Horizon	Class	Obj	Best Cost	Best GAP (%)	Avg Cost	Avg GAP (%)	Time (s)	Time (h)
LR1_DR08_VC10_V40b	120	M	206,315.00	214,510.52	3.82%	219,516.13	6.01%	74,077.05	20.58
LR1_DR12_VC05_V70a	120	H	278,647.00	273,841.24	-1.75%	278,033.42	-0.22%	85,925.81	23.87
LR1_DR12_VC05_V70b	120	M	308,555.00	307,224.68	-0.43%	317,056.67	2.68%	86,596.32	24.05
LR1_DR12_VC10_V70a	120	H	283,154.00	280,455.19	-0.96%	285,766.15	0.91%	86,622.00	24.06
LR1_DR12_VC10_V70b	120	H	295,126.00	290,283.87	-1.67%	295,120.22	0.00%	86,632.19	24.06

Table D.23

MIRPPacs results for instance with horizon = 180.

Instance	Horizon	Class	Obj	Best Cost	Best GAP (%)	Avg Cost	Avg GAP (%)	Time (s)	Time (h)
LR1_DR02_VC01_V6a	180	E	52,167.00	52,167.21	0.00%	52,167.21	0.00%	2,011.79	0.56
LR1_DR02_VC02_V6a	180	E	129,372.00	131,528.86	1.64%	132,299.84	2.21%	2,888.81	0.80
LR1_DR02_VC03_V7a	180	M	60,547.00	60,604.56	0.09%	66,913.85	9.51%	16,536.11	4.59
LR1_DR02_VC03_V8a	180	E	68,153.00	67,480.80	-1.00%	74,564.91	8.60%	12,257.40	3.40
LR1_DR02_VC04_V8a	180	M	66,017.00	66,135.36	0.18%	66,557.38	0.81%	7,945.02	2.21
LR1_DR02_VC05_V8a	180	M	58,619.00	58,448.57	-0.29%	58,989.91	0.63%	7,989.09	2.22
LR1_DR02_VC03_V10b	180	M	125,638.00	122,430.23	-2.62%	132,403.04	5.11%	21,784.37	6.05
LR1_DR03_VC03_V13b	180	E	165,764.00	159,578.98	-3.88%	165,490.72	-0.17%	46,937.43	13.04
LR1_DR03_VC03_V16a	180	M	143,178.00	151,332.88	5.39%	160,799.28	10.96%	14,352.34	3.99
LR1_DR04_VC03_V15a	180	M	118,621.00	118,086.24	-0.45%	119,037.85	0.35%	34,908.87	9.70
LR1_DR04_VC03_V15b	180	E	189,989.00	184,580.68	-2.93%	190,540.39	0.29%	84,781.38	23.55
LR1_DR04_VC05_V17a	180	M	117,710.00	117,021.59	-0.59%	117,686.22	-0.02%	58,026.39	16.12
LR1_DR04_VC05_V17b	180	M	159,168.00	152,808.19	-4.16%	158,141.43	-0.65%	77,057.28	21.40
LR1_DR05_VC05_V25a	180	M	171,620.00	170,012.90	-0.95%	182,850.48	6.14%	85,915.09	23.87
LR1_DR05_VC05_V25b	180	M	205,368.00	204,184.54	-0.58%	215,400.44	4.66%	86,251.32	23.96
LR1_DR08_VC05_V38a	180	H	274,244.00	270,114.44	-1.53%	284,603.35	3.64%	85,706.82	23.81
LR1_DR08_VC05_V40a	180	H	296,760.00	295,350.18	-0.48%	308,750.64	3.88%	86,575.42	24.05
LR1_DR08_VC05_V40b	180	M	337,559.00	332,554.16	-1.50%	358,776.56	5.91%	86,490.80	24.03
LR1_DR08_VC10_V40a	180	H	304,261.00	303,400.82	-0.28%	317,982.43	4.32%	86,521.60	24.03
LR1_DR08_VC10_V40b	180	H	331,775.00	339,360.88	2.24%	357,073.60	7.08%	82,951.25	23.04
LR1_DR12_VC05_V70a	180	H	460,566.00	467,057.26	1.39%	502,877.62	8.41%	86,600.84	24.06
LR1_DR12_VC05_V70b	180	H	491,160.00	503,340.69	2.42%	546,459.67	10.12%	86,500.11	24.03
LR1_DR12_VC10_V70a	180	H	466,975.00	483,946.03	3.51%	507,494.51	7.98%	86,665.53	24.07
LR1_DR12_VC10_V70b	180	H	470,172.00	478,275.23	1.69%	506,508.75	7.17%	86,679.11	24.08

Table D.24

MIRPPacs results for instance with horizon = 360.

Instance	Horizon	Class	Obj	Best Cost	Best GAP (%)	Avg Cost	Avg GAP (%)	Time (s)	Time (h)
LR1_DR02_VC01_V6a	360	E	108,141.00	108,141.02	0.00%	108,141.02	0.00%	4,112.34	1.14
LR1_DR02_VC02_V6a	360	E	283,031.00	281,837.72	-0.42%	289,796.45	2.33%	18,231.62	5.06
LR1_DR02_VC03_V7a	360	H	124,282.00	135,418.13	8.22%	161,475.33	23.03%	26,731.64	7.43
LR1_DR02_VC03_V8a	360	M	141,166.00	152,220.62	7.26%	166,783.19	15.36%	38,021.60	10.56
LR1_DR02_VC04_V8a	360	M	138,693.00	140,110.94	1.01%	148,069.03	6.33%	66,674.24	18.52
LR1_DR02_VC05_V8a	360	H	122,598.00	122,845.90	0.20%	136,288.95	10.05%	33,431.84	9.29
LR1_DR03_VC03_V10b	360	H	259,888.00	274,635.46	5.37%	302,560.78	14.10%	43,086.27	11.97
LR1_DR03_VC03_V13b	360	M	316,441.00	319,851.08	1.07%	330,697.62	4.31%	66,357.99	18.43
LR1_DR03_VC03_V16a	360	M	327,793.00	387,079.30	15.32%	424,330.22	22.75%	38,343.39	10.65
LR1_DR04_VC03_V15a	360	M	252,710.00	253,436.60	0.29%	269,180.18	6.12%	62,519.59	17.37
LR1_DR04_VC03_V15b	360	M	339,308.00	363,832.68	6.74%	388,056.28	12.56%	81,291.53	22.58
LR1_DR04_VC05_V17a	360	H	251,623.00	252,753.92	0.45%	270,717.98	7.05%	80,996.66	22.50
LR1_DR04_VC05_V17b	360	M	304,507.00	316,388.38	3.76%	340,896.16	10.67%	83,273.25	23.13
LR1_DR05_VC05_V25a	360	H	368,628.00	419,634.16	12.15%	1,294,438.54	71.52%	73,762.74	20.49
LR1_DR05_VC05_V25b	360	H	410,053.00	444,752.59	7.80%	528,149.56	22.36%	86,570.27	24.05
LR1_DR08_VC05_V38a	360	H	596,969.00	643,104.53	7.17%	844,541.88	29.31%	86,516.59	24.03
LR1_DR08_VC05_V40a	360	H	652,380.00	906,012.71	27.99%	2,406,928.26	72.90%	70,942.03	19.71
LR1_DR08_VC05_V40b	360	H	709,713.00	922,880.44	23.10%	1,067,744.93	33.53%	86,510.90	24.03
LR1_DR08_VC10_V40a	360	H	663,245.00	786,274.69	15.65%	3,469,460.40	80.88%	70,202.71	19.50
LR1_DR08_VC10_V40b	360	H	724,513.00	913,262.05	20.67%	2,824,587.24	74.35%	66,803.09	18.56
LR1_DR12_VC05_V70a	360	H	1,021,389.00	1,154,456.04	11.53%	2,234,872.15	54.30%	86,585.16	24.05
LR1_DR12_VC05_V70b	360	H	1,093,013.00	1,295,507.38	15.63%	2,387,510.78	54.22%	86,438.75	24.01
LR1_DR12_VC10_V70a	360	H	1,024,399.00	1,202,266.42	14.79%	4,760,896.43	78.48%	76,175.67	21.16
LR1_DR12_VC10_V70b	360	H	1,001,541.00	1,263,001.69	20.70%	7,732,537.21	87.05%	82,194.82	22.83

Appendix C. Detailed results

This appendix provides the complete results from the computational experiments conducted with the proposed heuristic.

Tables C.16–C.18 show detailed results of the average total cost and time are shown for each instance, in three different time horizons for different values of N . It is possible to verify the convergence behavior within each stage of the algorithm and the trade-offs between quality and computational effort.

Tables C.19–C.21 present the best-performing results identified among all tested values of N .

Appendix D. Complete MIRPPacs results

This appendix provides the complete results from the computational experiments conducted with the MIRPPacs (Munguía et al., 2019) running in 16 threads. Detailed results of the best total cost, the best

GAP (%), the average total cost, the average GAP (%), and time are shown for each instance (see Tables D.22–D.24).

Data availability

The data is public. The code is available on request.

References

- Agra, A., Christiansen, M., Delgado, A., Simonetti, L., 2014. Hybrid heuristics for a short sea inventory routing problem. *European J. Oper. Res.* 236 (3), 924–935.
- Agra, A., Christiansen, M., Hvattum, L.M., Rodrigues, F., 2016. A MIP based local search heuristic for a stochastic maritime inventory routing problem. In: *International Conference on Computational Logistics*. Springer, pp. 18–34.
- Agra, A., Christiansen, M., Hvattum, L.M., Rodrigues, F., 2018. Robust optimization for a maritime inventory routing problem. *Transp. Sci.* 52 (3), 509–525.
- Agra, A., Christiansen, M., Ivarsoy, K.S., Solhaug, I.E., Tomasgard, A., 2017. Combined ship routing and inventory management in the salmon farming industry. *Ann. Oper. Res.* 253, 799–823.
- Al-Haidous, S., Govindan, R., Elomri, A., Al-Ansari, T., 2022. An optimization approach to increasing sustainability and enhancing resilience against environmental constraints in LNG supply chains: A Qatar case study. *Energy Rep.* 8, 9742–9756.
- Andersson, H., Hoff, A., Christiansen, M., Hasle, G., Løkketangen, A., 2010. Industrial aspects and literature survey: Combined inventory management and routing. *Comput. Oper. Res.* 37 (9), 1515–1536.
- Chandra, S., Srivastava, R.K., Agarwal, Y., 2013. Multi-product maritime inventory routing with optional cargoes: An application to outbound automotive logistics. *J. Adv. Manag. Res.* 10 (2), 206–229.
- Christiansen, M., Fagerholt, K., 2009. Maritime inventory routing problems. *Encycl. Optim.* 2, 1947–1955.
- Christiansen, M., Fagerholt, K., Flatberg, T., Haugen, Ø., Kloster, O., Lund, E.H., 2011. Maritime inventory routing with multiple products: A case study from the cement industry. *European J. Oper. Res.* 208 (1), 86–94.
- da Costa, L.G.V., 2018. *Matheurísticas Para a Roteirização De Navios Com Estoques E Múltiplos Produtos* (Master's thesis). PUC-Rio, In Portuguese.
- De, A., Kumar, S.K., Gunasekaran, A., Tiwari, M.K., 2017. Sustainable maritime inventory routing problem with time window constraints. *Eng. Appl. Artif. Intell.* 61, 77–95.
- de Dios, J.-A.M., Mezura-Montes, E., Quiroz-Castellanos, M., 2021. Automated parameter tuning as a bilevel optimization problem solved by a surrogate-assisted population-based approach. *Appl. Intell.* 51 (8), 5978–6000.
- Diz, G.S.d.S., Hamacher, S., Oliveira, F., 2019. A robust optimization model for the maritime inventory routing problem. *Flex. Serv. Manuf. J.* 31, 675–701.
- Dong, Y., Maravelias, C.T., Jerome, N.F., 2018. Reoptimization framework and policy analysis for maritime inventory routing under uncertainty. *Optim. Eng.* 19, 937–976.
- Fagerholt, K., Hvattum, L.M., Papageorgiou, D.J., Urrutia, S., 2023. Maritime inventory routing: recent trends and future directions. *Int. Trans. Oper. Res.* 30, 3013–3056.
- Friske, M.W., Buriol, L.S., 2020. A multi-start algorithm and a large neighborhood search for a maritime inventory routing problem. In: *2020 IEEE Congress on Evolutionary Computation. CEC, IEEE*, pp. 1–8.
- Friske, M.W., Buriol, L.S., Camponogara, E., 2022. A relax-and-fix and fix-and-optimize algorithm for a Maritime Inventory Routing Problem. *Comput. Oper. Res.* 137, 105520.
- Goel, V., Slusky, M., van Hoeve, W.-J., Furman, K.C., Shao, Y., 2015. Constraint programming for LNG ship scheduling and inventory management. *European J. Oper. Res.* 241 (3), 662–673.
- Hemmati, A., Hvattum, L.M., Christiansen, M., Laporte, G., 2016. An iterative two-phase hybrid matheuristic for a multi-product short sea inventory-routing problem. *European J. Oper. Res.* 252 (3), 775–788.
- Hemmati, A., Stålhane, M., Hvattum, L.M., Andersson, H., 2015. An effective heuristic for solving a combined cargo and inventory routing problem in tramp shipping. *Comput. Oper. Res.* 64, 274–282.
- Kirkpatrick, S., Gelatt, C.D., Vecchi, M.P., 1983. Optimization by simulated annealing. *Sci.* 220 (4598), 671–680.
- Li, M., Fagerholt, K., Schütz, P., 2023. Maritime inventory routing with transshipment: the case of Yamal LNG. *Flex. Serv. Manuf. J.* 35 (1), 269–294.
- Li, M., Schütz, P., 2020. Planning annual LNG deliveries with transshipment. *Energies* 13 (6), 1490.
- Lourenço, H.R., Martin, O.C., Stützle, T., 2003. Iterated local search. In: *Handbook of Metaheuristics*. Springer, pp. 320–353.
- Lowerre, B.T., 1976. *The Harpy Speech Recognition System* (Phd thesis). Carnegie Mellon University.
- Morais, R., Bulhões, T., Subramanian, A., 2024. Exact and heuristic algorithms for minimizing the makespan on a single machine scheduling problem with sequence-dependent setup times and release dates. *European J. Oper. Res.* 315 (2), 442–453.
- Msakni, M.K., Haouari, M., 2018. Short-term planning of liquefied natural gas deliveries. *Transp. Res. Part C: Emerg. Technol.* 90, 393–410.
- Munguía, L.-M., Ahmed, S., Bader, D.A., Nemhauser, G.L., Shao, Y., Papageorgiou, D.J., 2019. Tailoring parallel alternating criteria search for domain specific MIPs: Application to maritime inventory routing. *Comput. Oper. Res.* 111, 21–34.
- Mutlu, F., Msakni, M.K., Yildiz, H., Sönmez, E., Pokharel, S., 2016. A comprehensive annual delivery program for upstream liquefied natural gas supply chain. *European J. Oper. Res.* 250 (1), 120–130.
- Ow, P.S., Morton, T.E., 1988. Filtered beam search in scheduling†. *Int. J. Prod. Res.* 26 (1), 35–62.
- Papageorgiou, D.J., 2013. A mixed-integer linear programming formulation for “Group 2” maritime inventory routing instances.
- Papageorgiou, D.J., Cheon, M.-S., Harwood, S., Trespalacios, F., Nemhauser, G.L., 2018. Recent progress using matheuristics for strategic maritime inventory routing. In: *Modeling, Computing and Data Handling Methodologies for Maritime Transportation*. Springer, pp. 59–94.
- Papageorgiou, D.J., Nemhauser, G.L., Sokol, J., Cheon, M.-S., Keha, A.B., 2014. MIRPLib-A library of maritime inventory routing problem instances: Survey, core model, and benchmark results. *European J. Oper. Res.* 235 (2), 350–366.
- Reddy, R., 1977. *Speech Understanding Systems: A Summary of Results of the Five-Year Research Effort at Carnegie-Mellon University* (Ph.D. thesis). Carnegie-mellon university.
- Ronen, D., 2002. Marine inventory routing: Shipments planning. *J. Oper. Res. Soc.* 53 (1), 108–114.
- Sanghikian, N., Martinelli, R., Abu-Marrul, V., 2021. A hybrid VNS for the multi-product maritime inventory routing problem. In: *International Conference on Variable Neighborhood Search*. Springer, pp. 111–122.
- Shaabani, H., Hoff, A., Hvattum, L.M., Laporte, G., 2023. A matheuristic for the multi-product maritime inventory routing problem. *Comput. Oper. Res.* 154, 106214.
- Shao, Y., Furman, K.C., Goel, V., Hoda, S., 2015. A hybrid heuristic strategy for liquefied natural gas inventory routing. *Transp. Res. Part C: Emerg. Technol.* 53, 151–171.
- Sheikhtajian, S., Nazemi, A., Feshari, M., 2020. Marine inventory-routing problem for liquefied natural gas under travel time uncertainty. *Int. J. Supply Oper. Manag.* 7 (1), 93–111.
- Siswanto, N., Essam, D., Sarker, R., 2011a. Multi-heuristics based genetic algorithm for solving maritime inventory routing problem. In: *2011 IEEE International Conference on Industrial Engineering and Engineering Management*. IEEE, pp. 116–120.
- Siswanto, N., Essam, D., Sarker, R., 2011b. Solving the ship inventory routing and scheduling problem with undedicated compartments. *Comput. Ind. Eng.* 61 (2), 289–299.
- Siswanto, N., Wiratno, S.E., Rusdiansyah, A., Sarker, R., 2019. Maritime inventory routing problem with multiple time windows. *J. Ind. Manag. Optim.* 15 (3), 1185.
- Song, J.-H., Furman, K.C., 2013. A maritime inventory routing problem: Practical approach. *Comput. Oper. Res.* 40 (3), 657–665.
- Stålhane, M., Rakke, J.G., Moe, C.R., Andersson, H., Christiansen, M., Fagerholt, K., 2012. A construction and improvement heuristic for a liquefied natural gas inventory routing problem. *Comput. Ind. Eng.* 62 (1), 245–255.
- Stanzani, A.d.L., Pureza, V., Morabito, R., Silva, B.J.V.d., Yamashita, D., Ribas, P.C., 2018. Optimizing multiship routing and scheduling with constraints on inventory levels in a Brazilian oil company. *Int. Trans. Oper. Res.* 25 (4), 1163–1198.
- Subramanian, A., Drummond, L.M.A., Bentes, C., Ochi, L.S., Farias, R., 2010. A parallel heuristic for the vehicle routing problem with simultaneous pickup and delivery. *Comput. Oper. Res.* 37 (11), 1899–1911.
- Uggen, K.T., Fodstad, M., Nørstebø, V.S., 2013. Using and extending fix-and-relax to solve maritime inventory routing problems. *Top* 21 (2), 355–377.
- Wang, Y., Wang, N., Han, P., 2023. Maritime location inventory routing problem for island supply chain network under periodic freight demand. *Comput. Oper. Res.* 149, 106042.
- Zhang, C., Nemhauser, G., Sokol, J., Cheon, M.-S., Keha, A., 2018. Flexible solutions to maritime inventory routing problems with delivery time windows. *Comput. Oper. Res.* 89, 153–162.